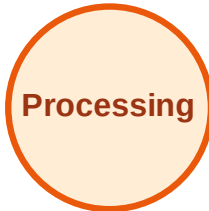


BFS Traversal

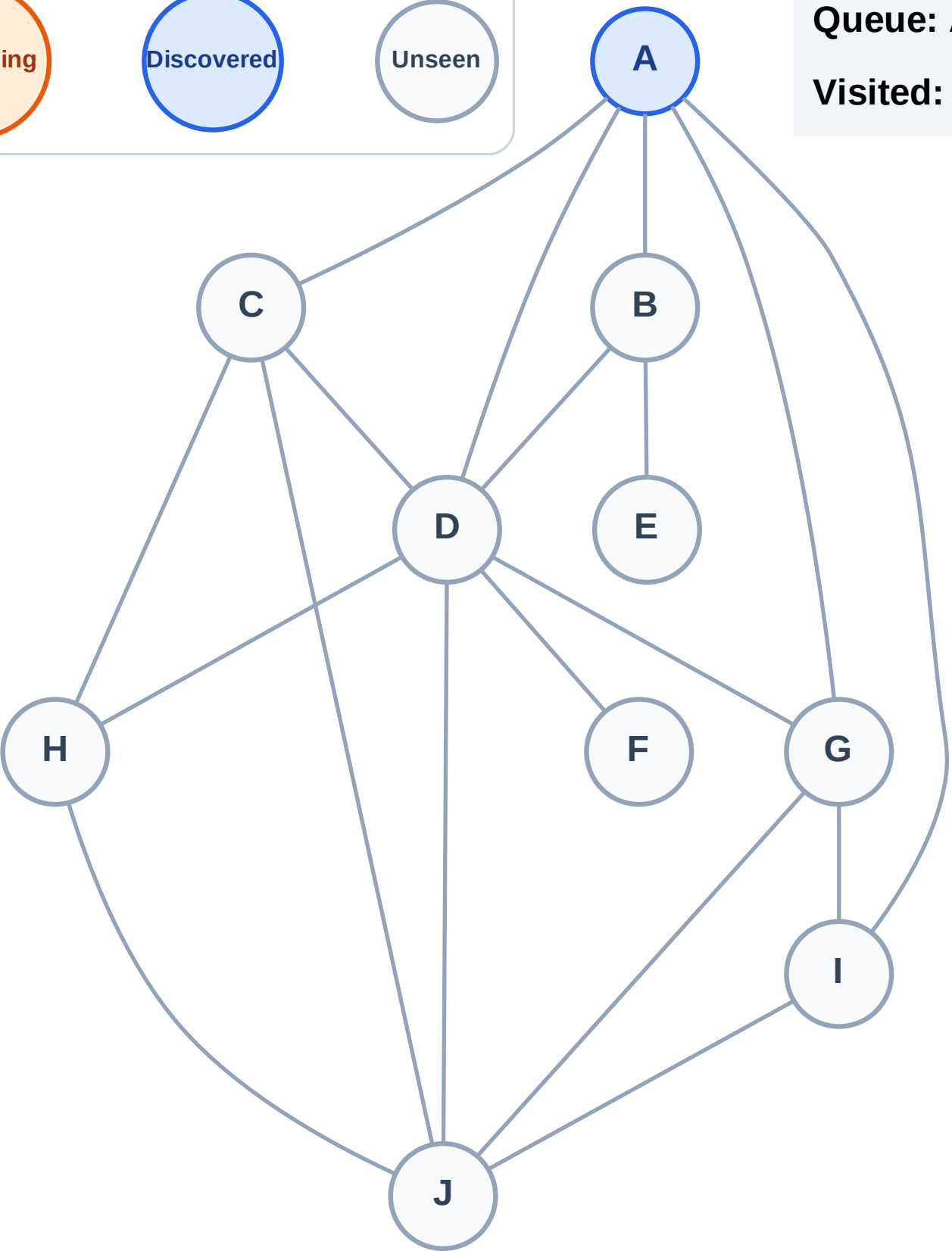
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

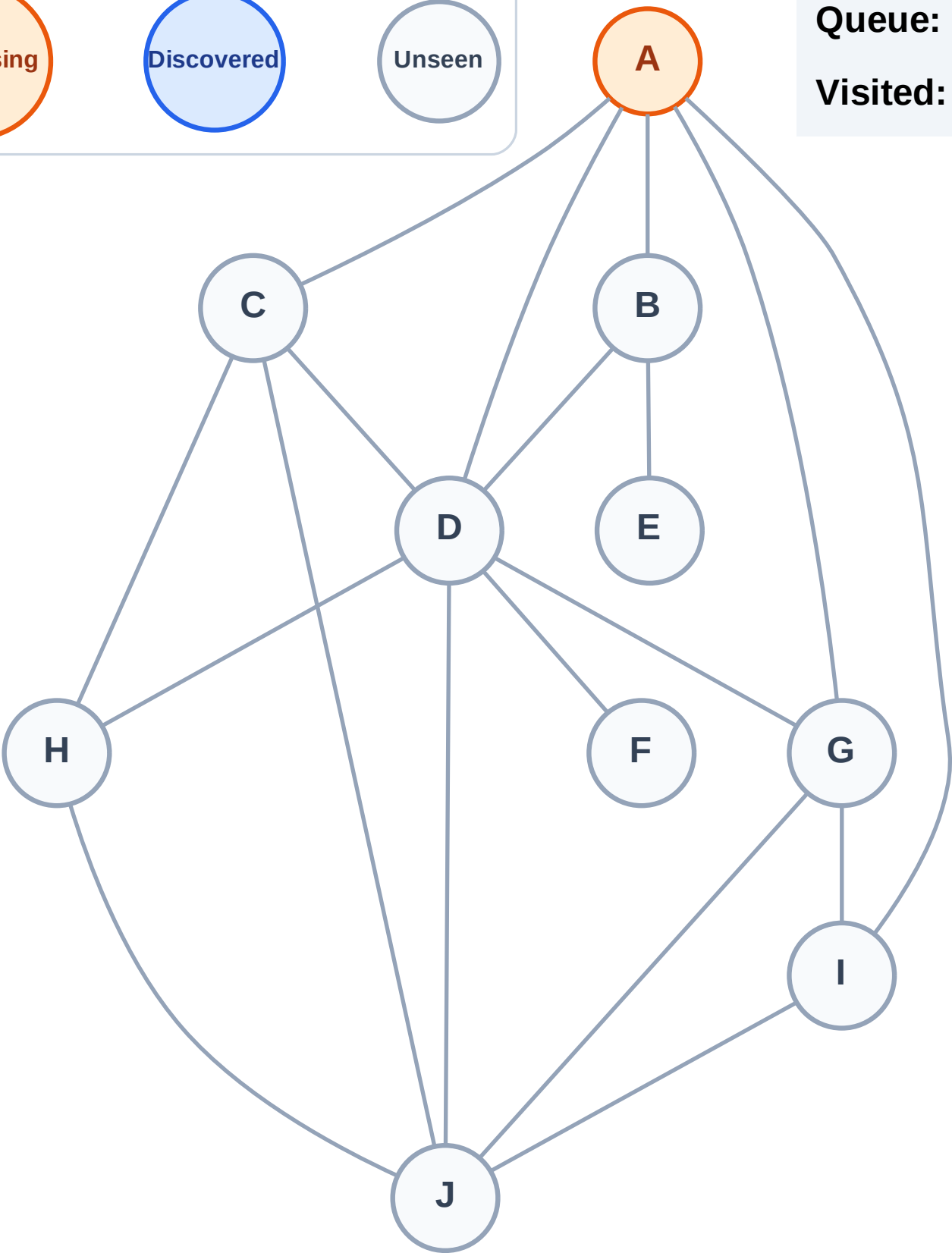
Step 2: Process node A

Legend



Queue: (empty)

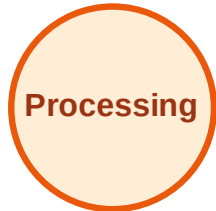
Visited: (none)



BFS Traversal

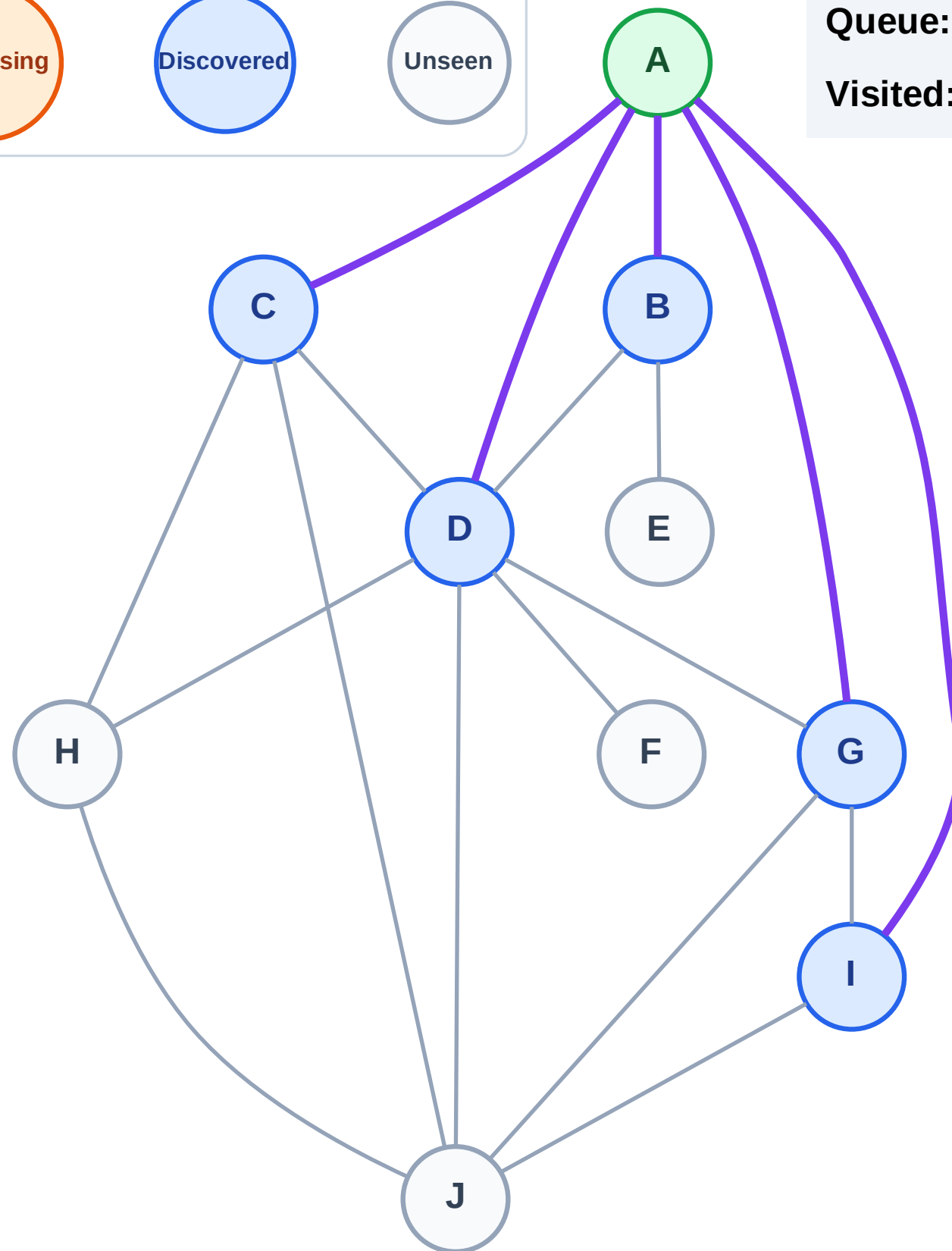
Step 3: Discover B, C, D, G, I from A

Legend



Queue: B → C → D → G → I

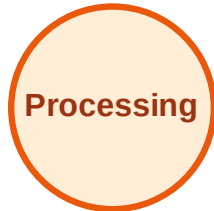
Visited: A



BFS Traversal

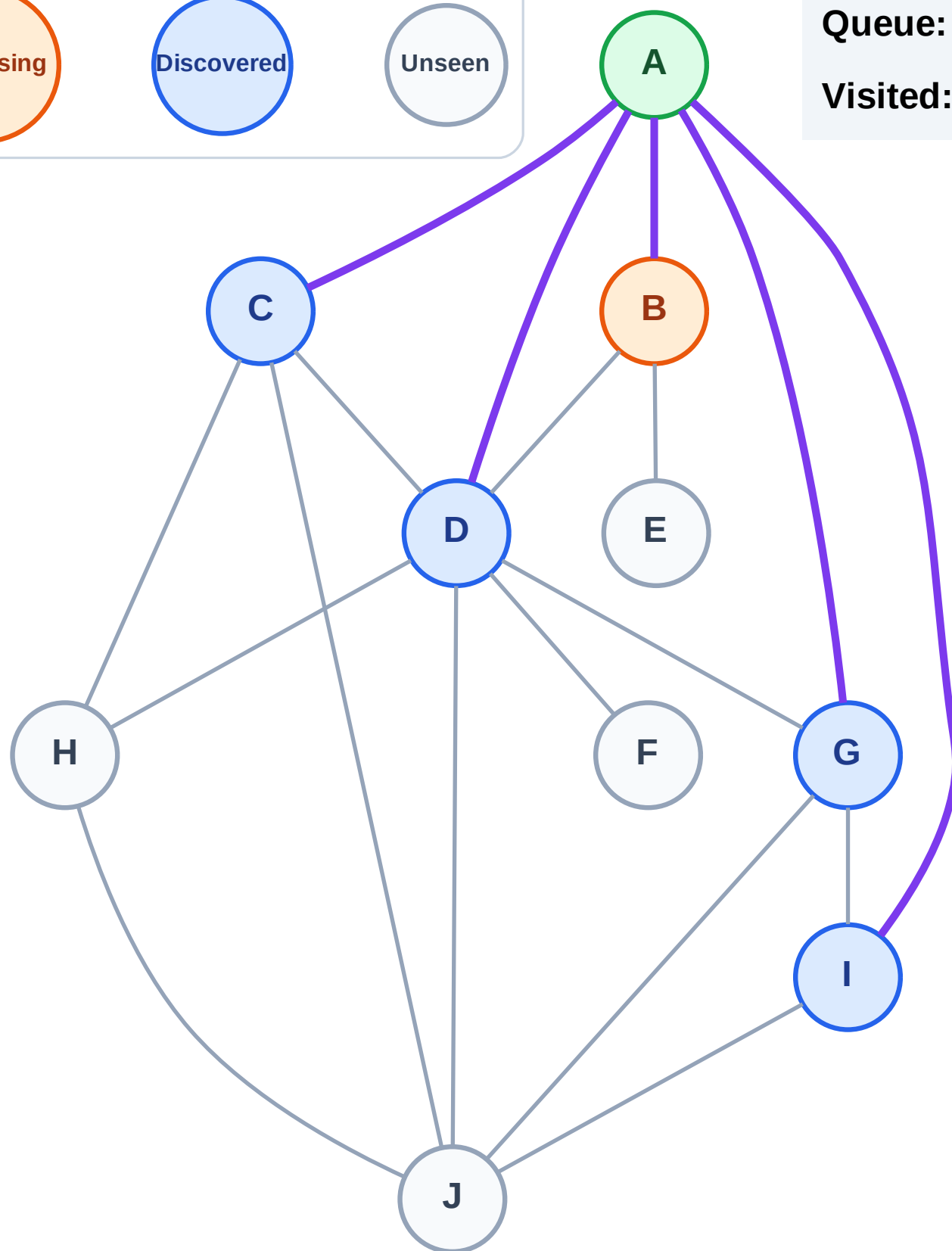
Step 4: Process node B

Legend



Queue: C → D → G → I

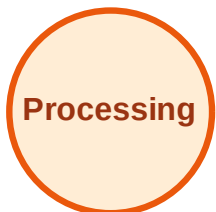
Visited: A



BFS Traversal

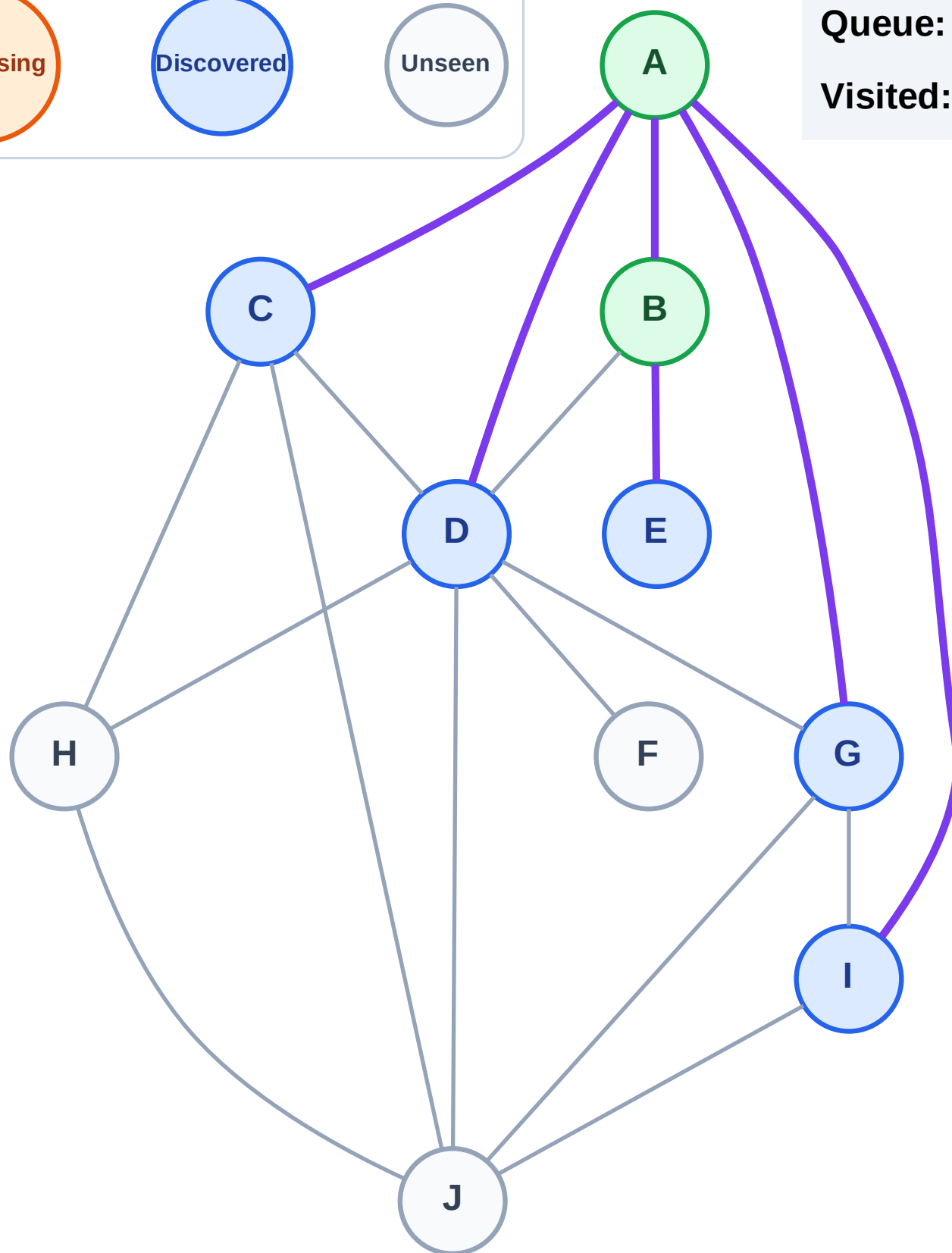
Step 5: Discover E from B

Legend



Queue: C → D → G → I → E

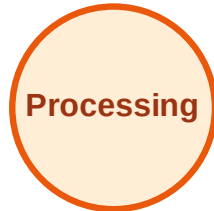
Visited: A, B



BFS Traversal

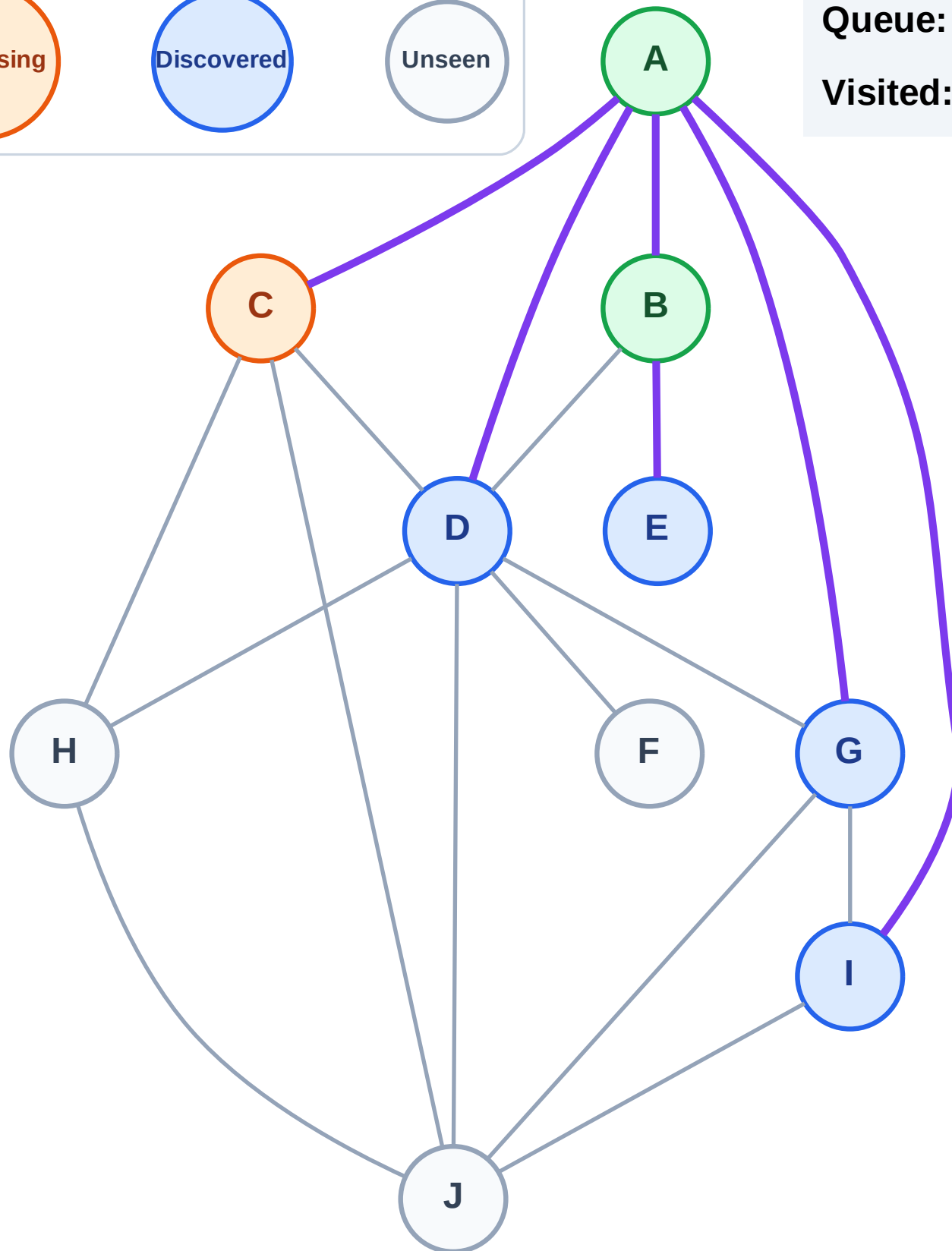
Step 6: Process node C

Legend



Queue: D → G → I → E

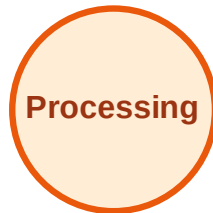
Visited: A, B



BFS Traversal

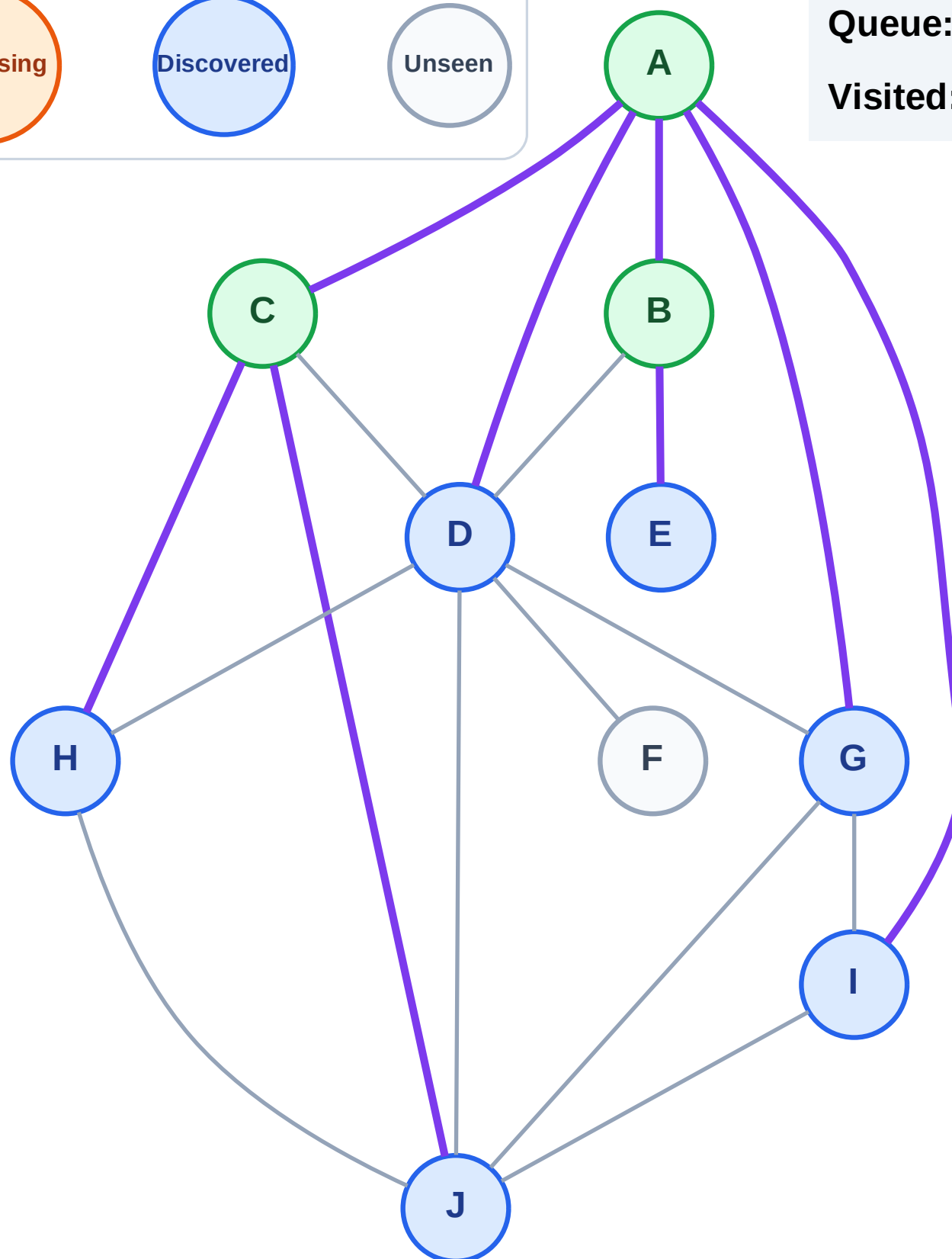
Step 7: Discover H, J from C

Legend



Queue: D → G → I → E → H → J

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

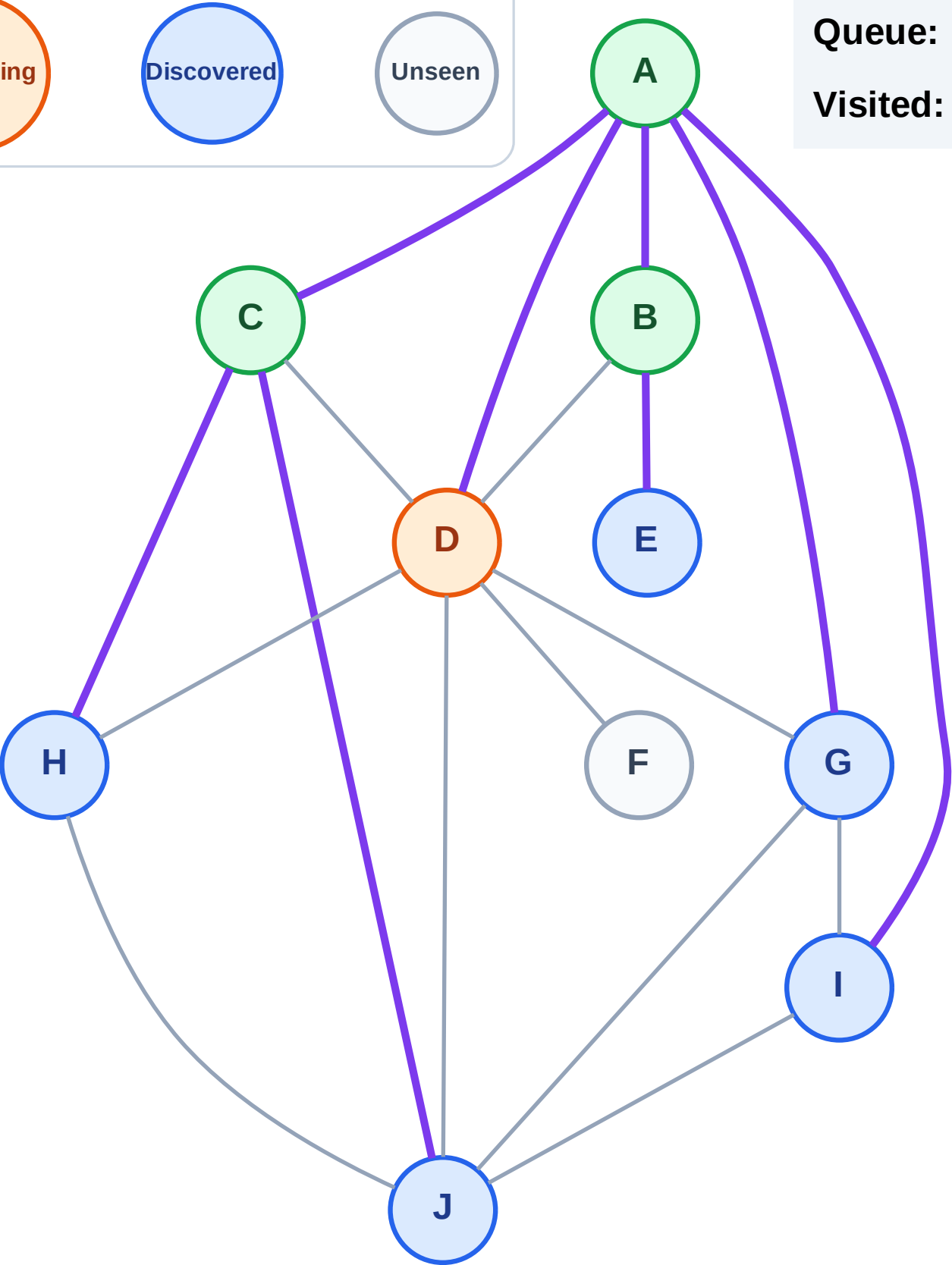
Processing

Discovered

Unseen

Queue: G → I → E → H → J

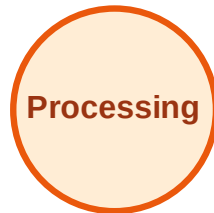
Visited: A, B, C



BFS Traversal

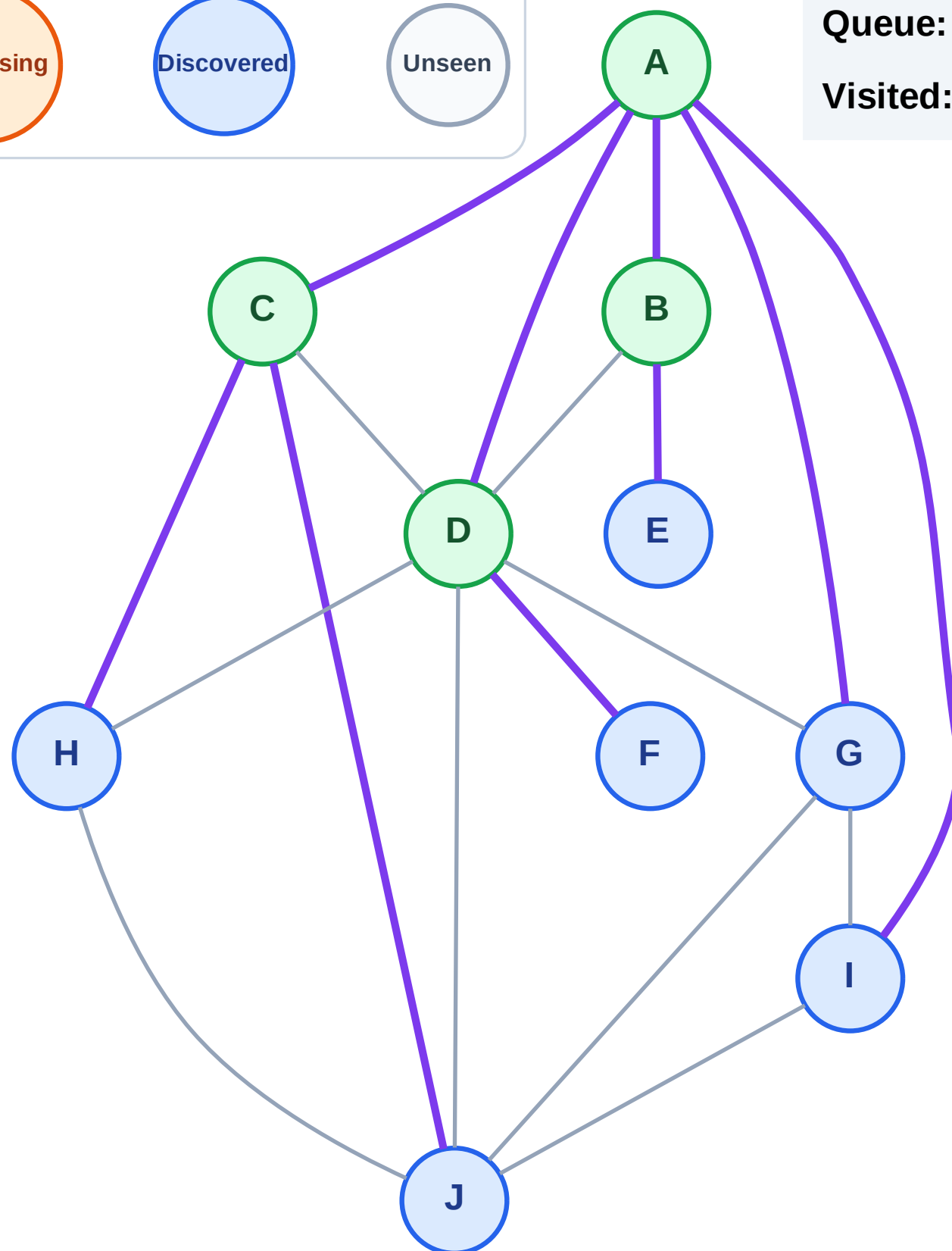
Step 9: Discover F from D

Legend



Queue: G → I → E → H → J → F

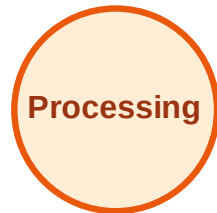
Visited: A, B, C, D



BFS Traversal

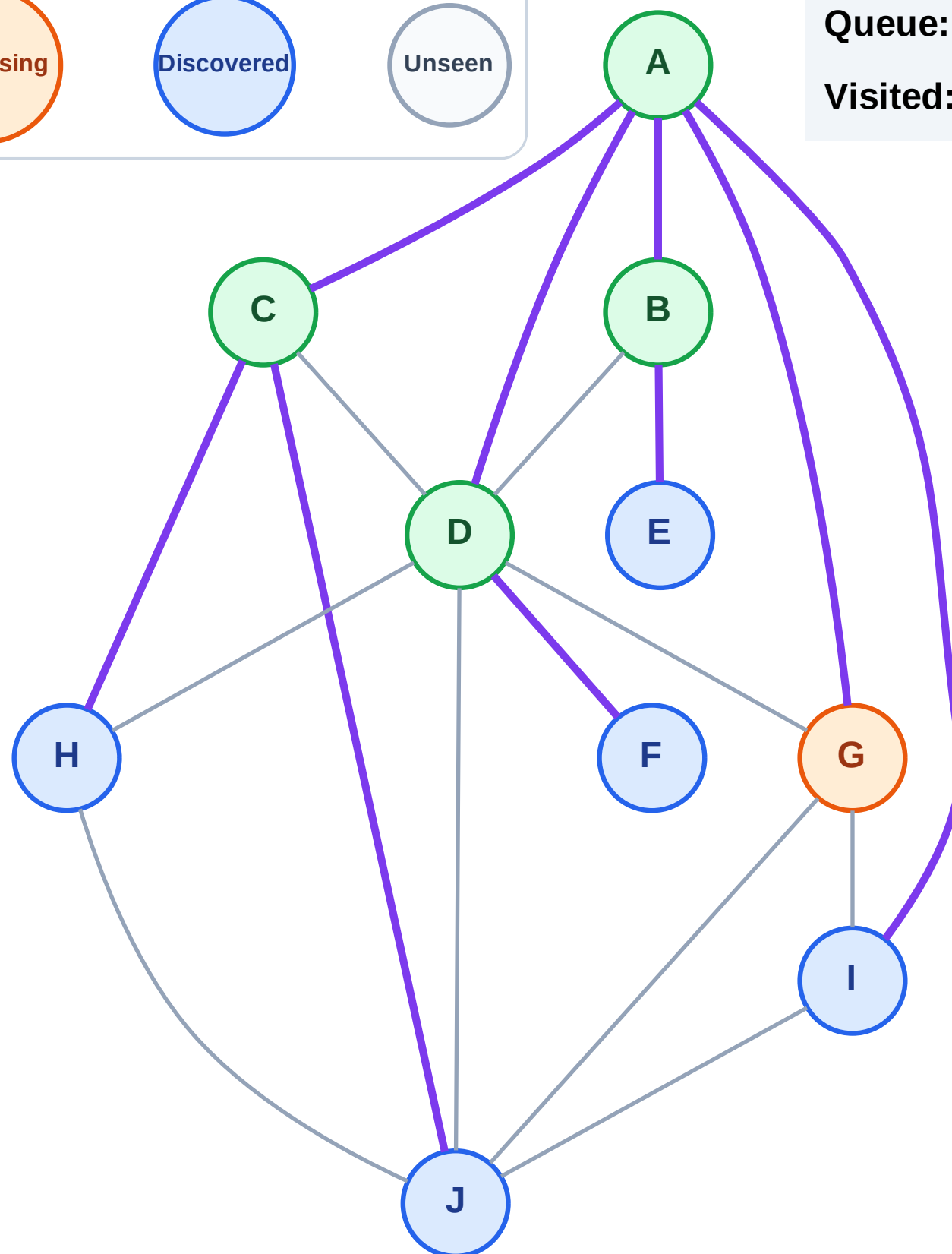
Step 10: Process node G

Legend



Queue: I → E → H → J → F

Visited: A, B, C, D



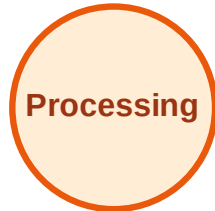
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



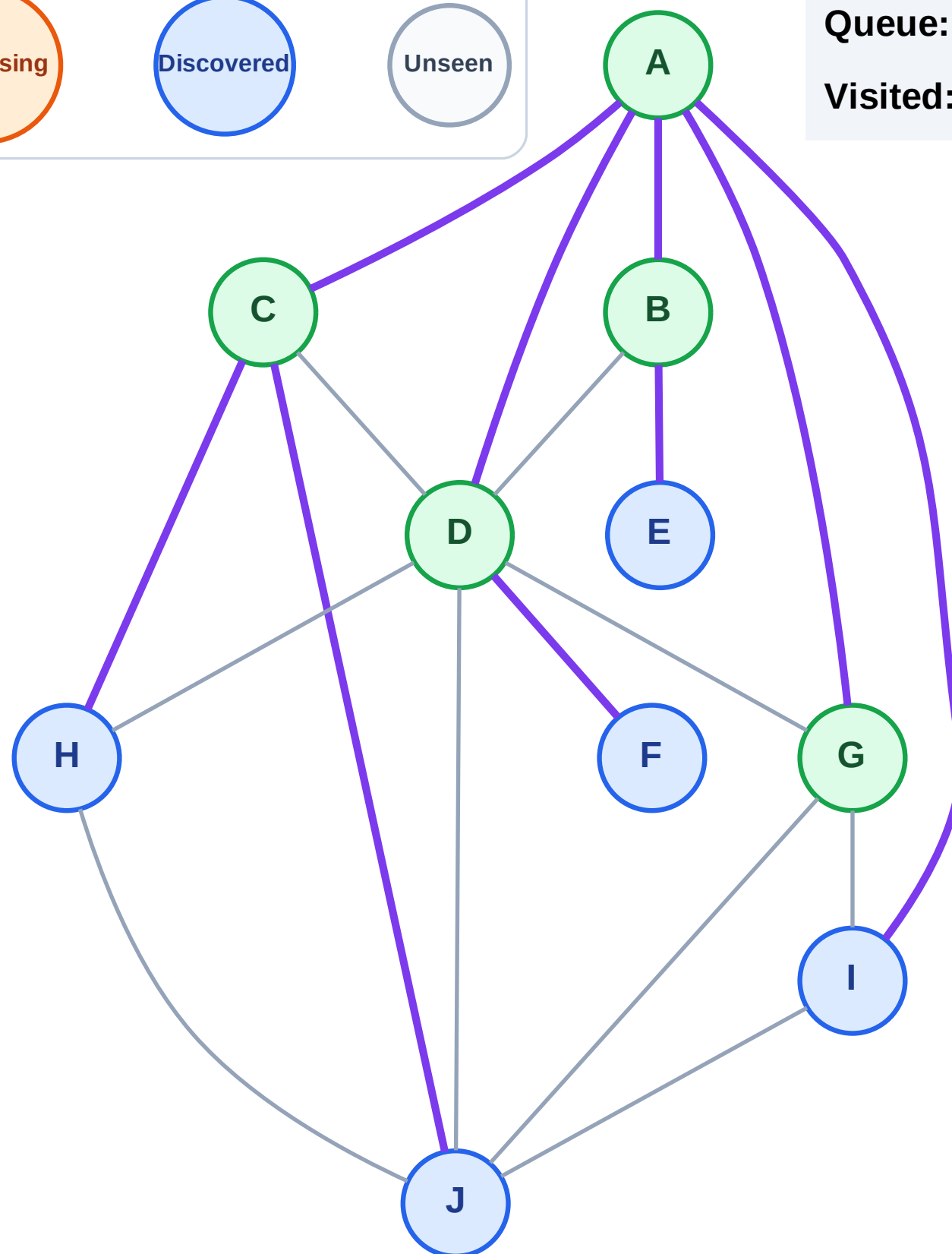
Discovered



Unseen

Queue: I → E → H → J → F

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node I

Legend

Complete

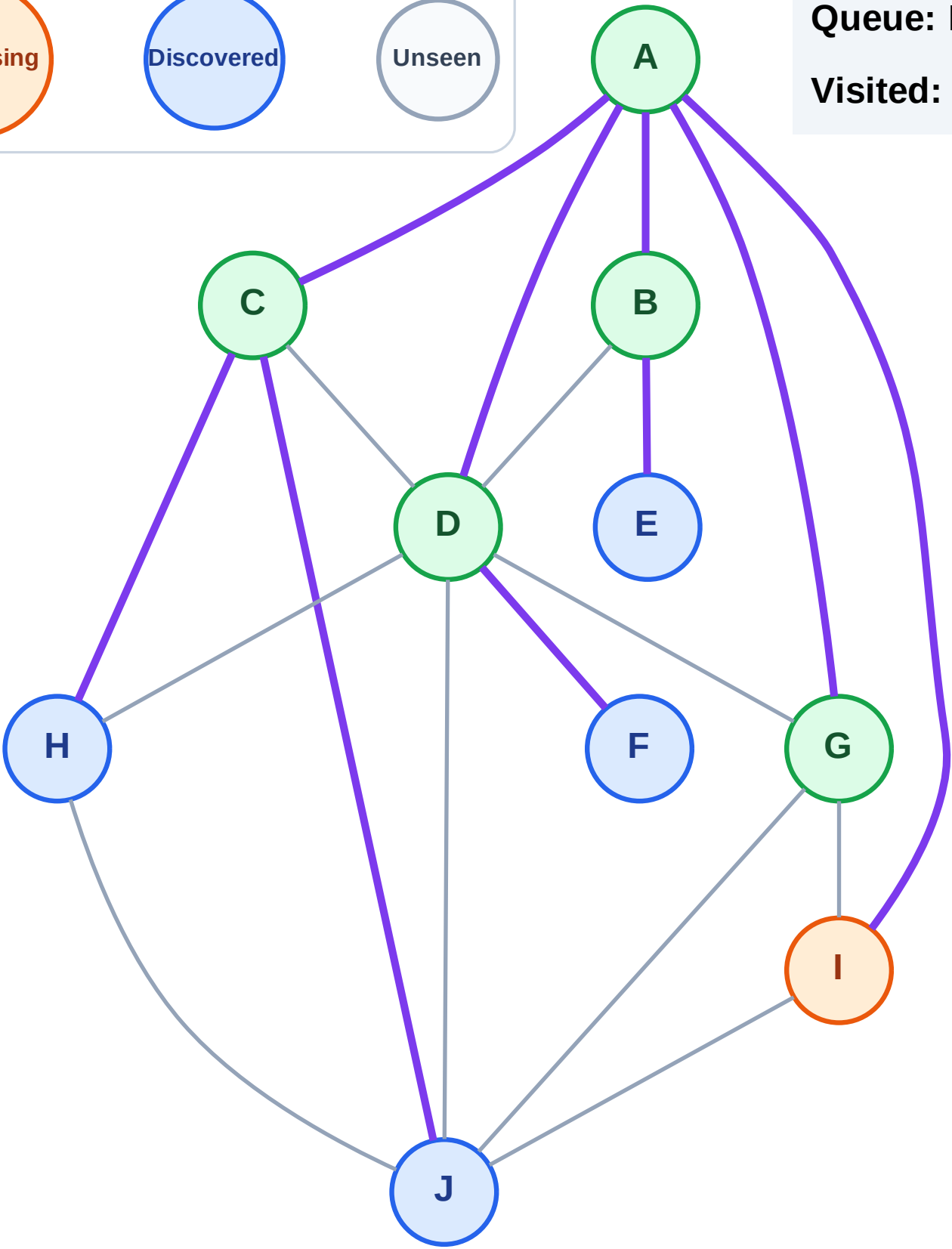
Processing

Discovered

Unseen

Queue: E → H → J → F

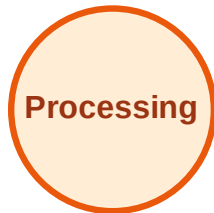
Visited: A, B, C, D, G



BFS Traversal

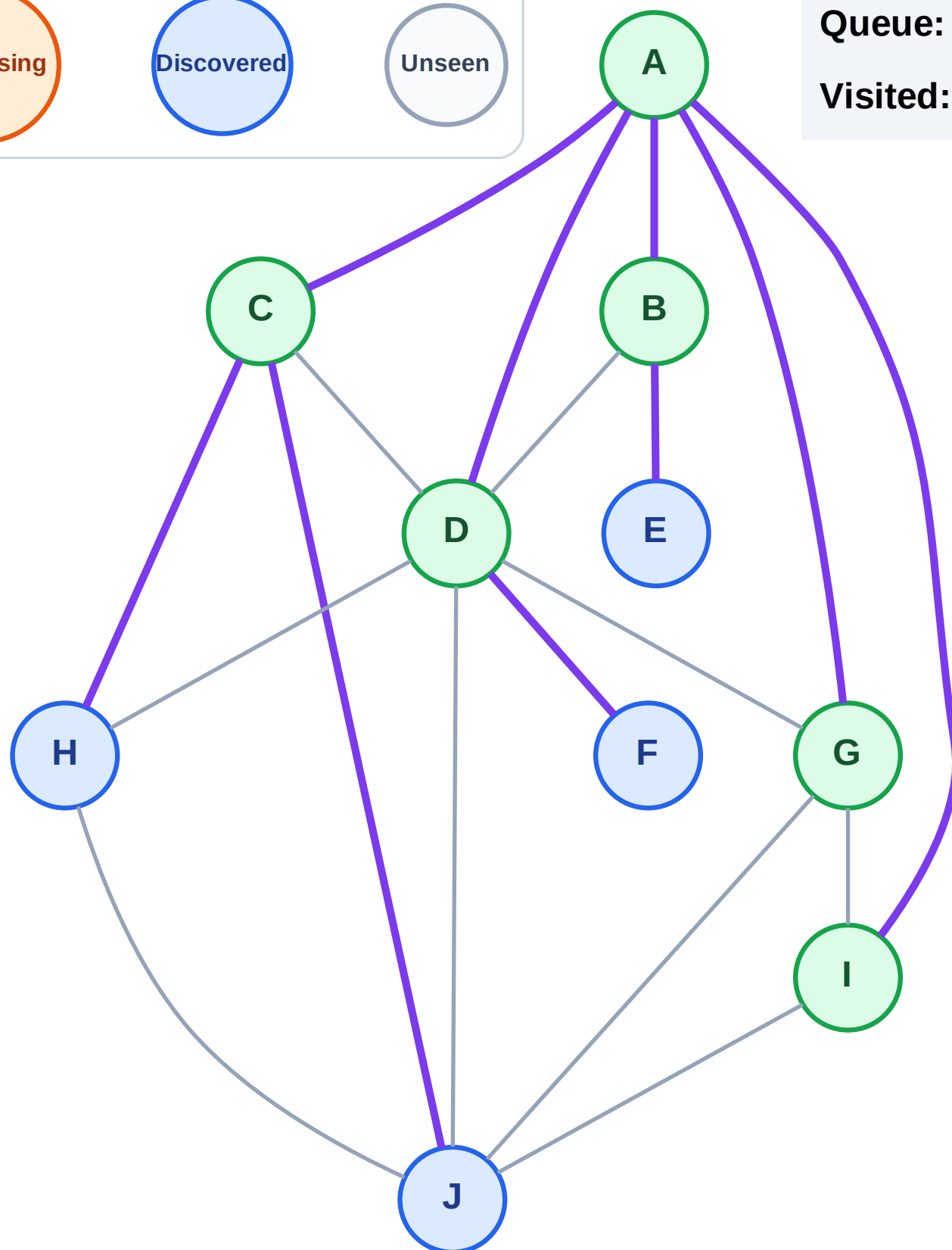
Step 13: No new neighbors from I

Legend



Queue: E → H → J → F

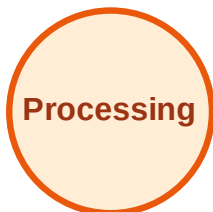
Visited: A, B, C, D, G, I



BFS Traversal

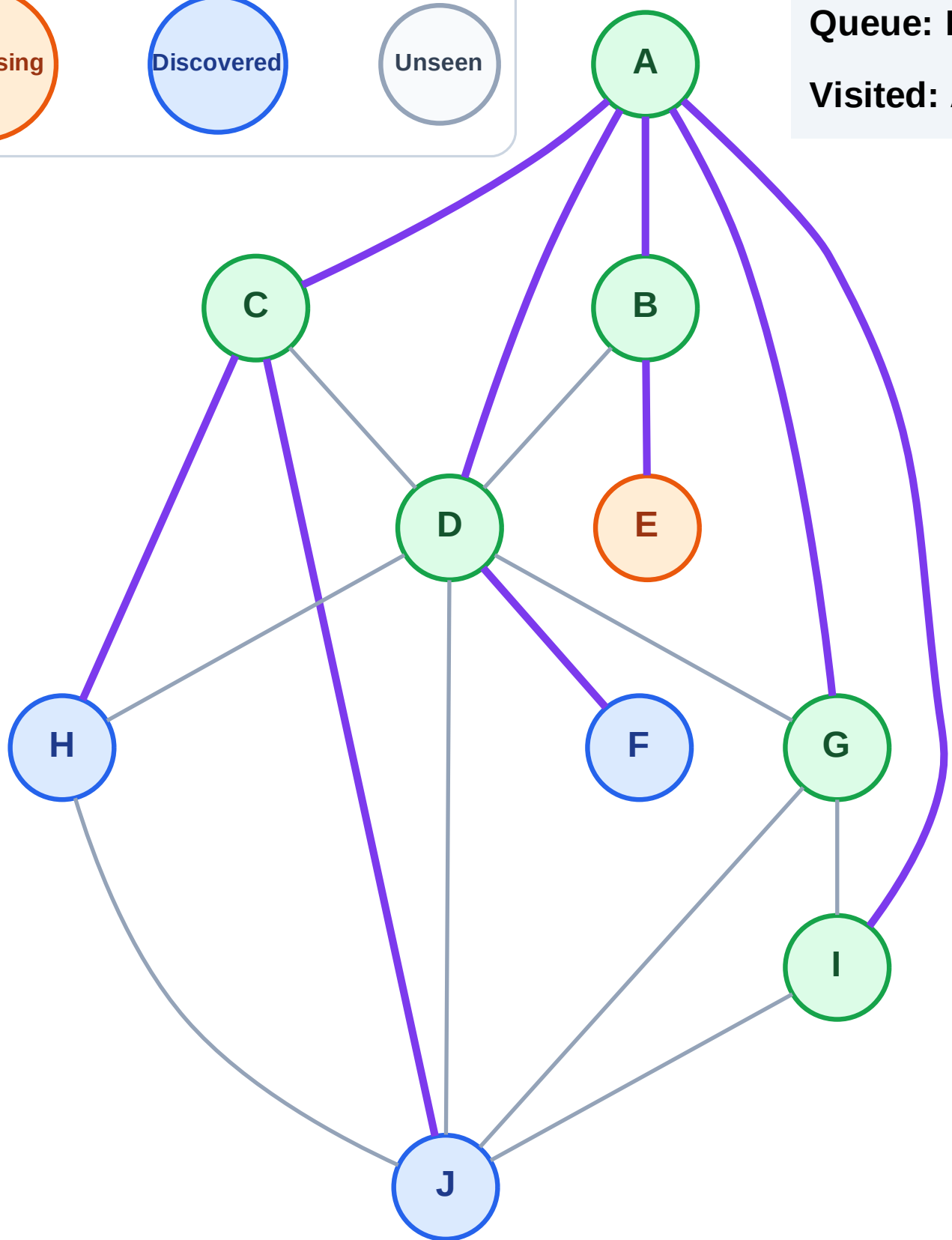
Step 14: Process node E

Legend



Queue: H → J → F

Visited: A, B, C, D, G, I



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

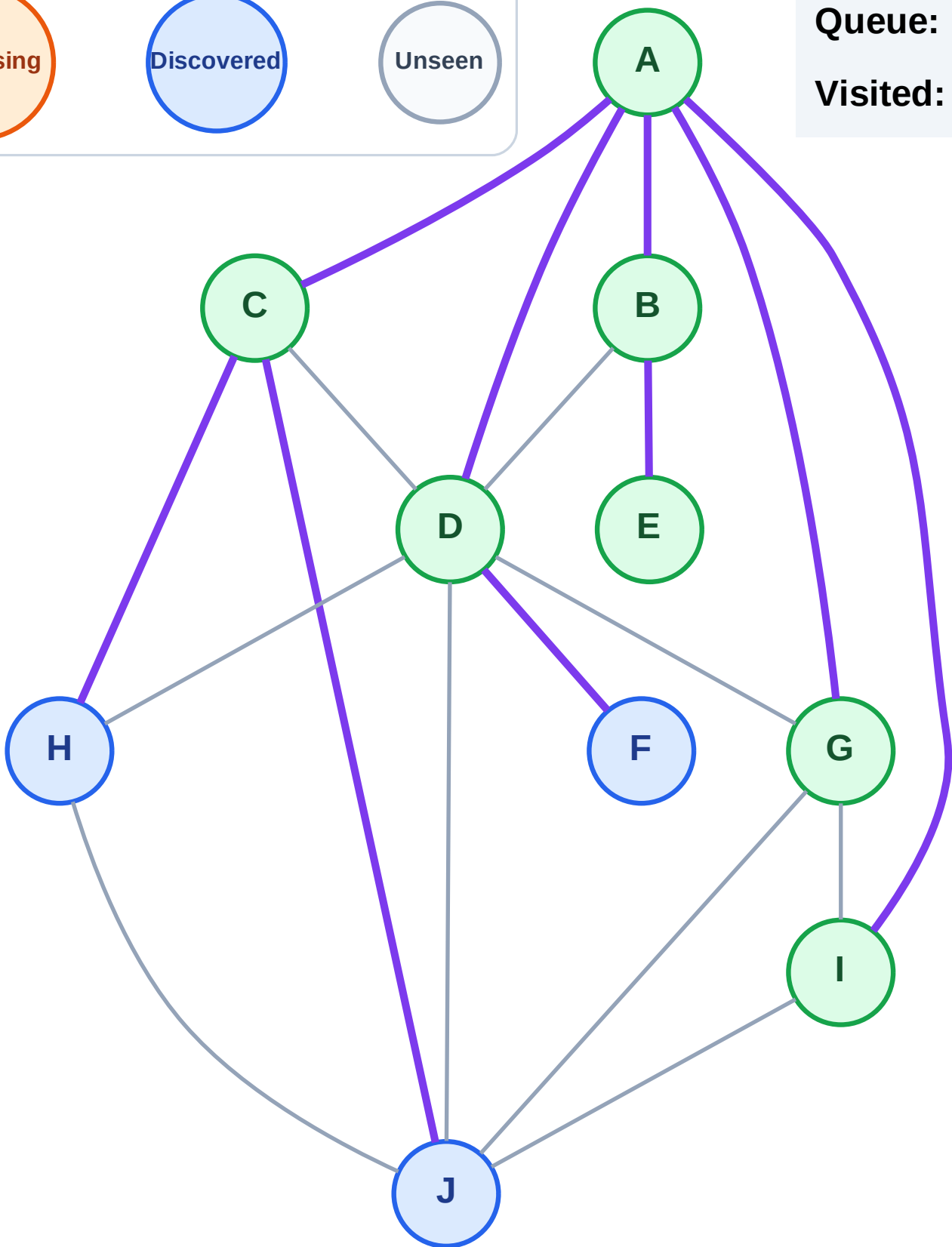
Processing

Discovered

Unseen

Queue: H → J → F

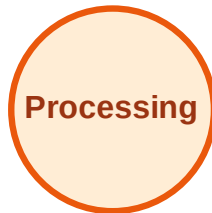
Visited: A, B, C, D, E, G, I



BFS Traversal

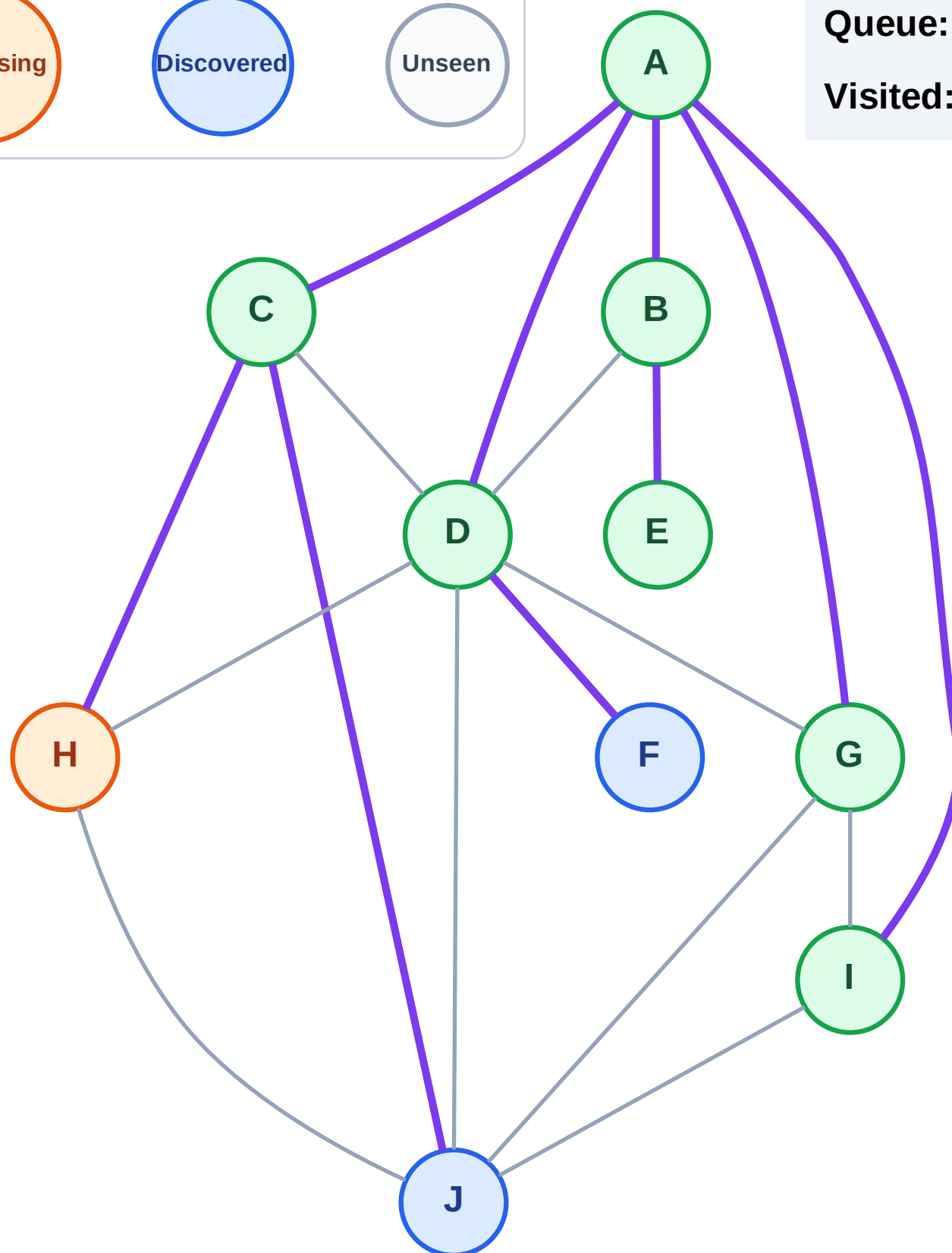
Step 16: Process node H

Legend



Queue: J → F

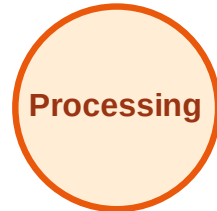
Visited: A, B, C, D, E, G, I



BFS Traversal

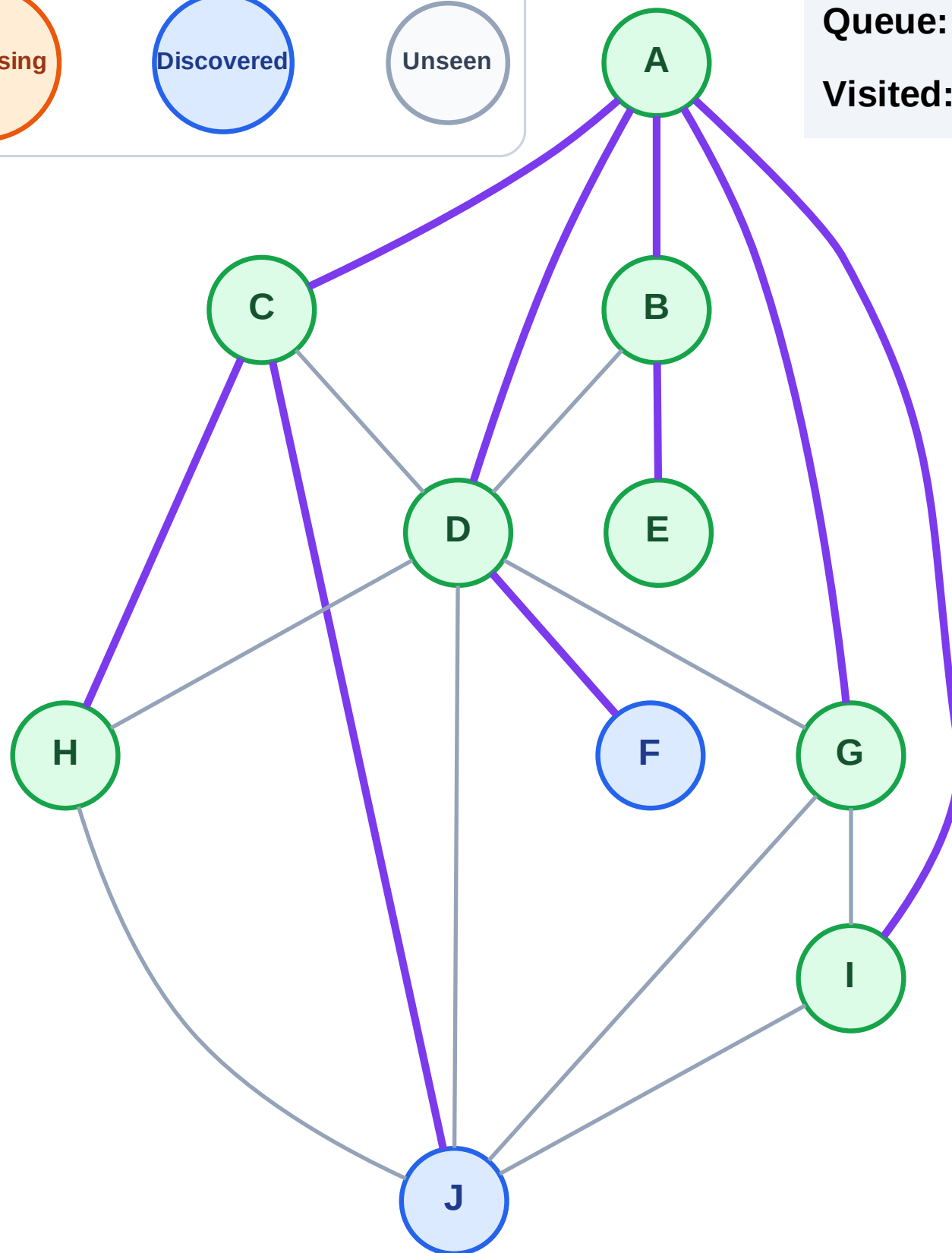
Step 17: No new neighbors from H

Legend



Queue: J → F

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node J

Legend

Complete

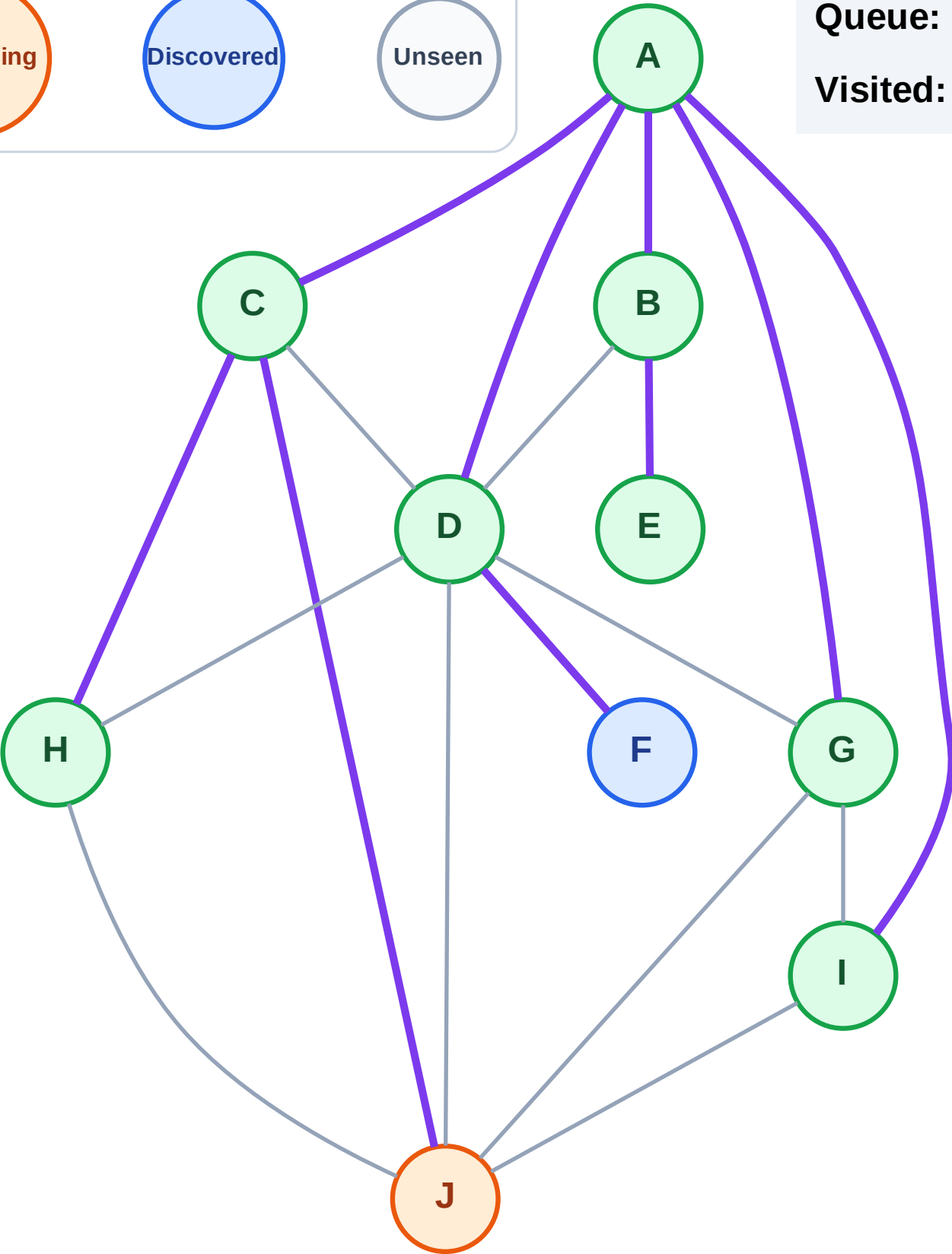
Processing

Discovered

Unseen

Queue: F

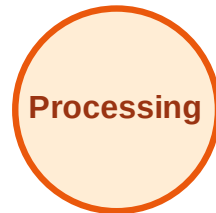
Visited: A, B, C, D, E, G, H, I



BFS Traversal

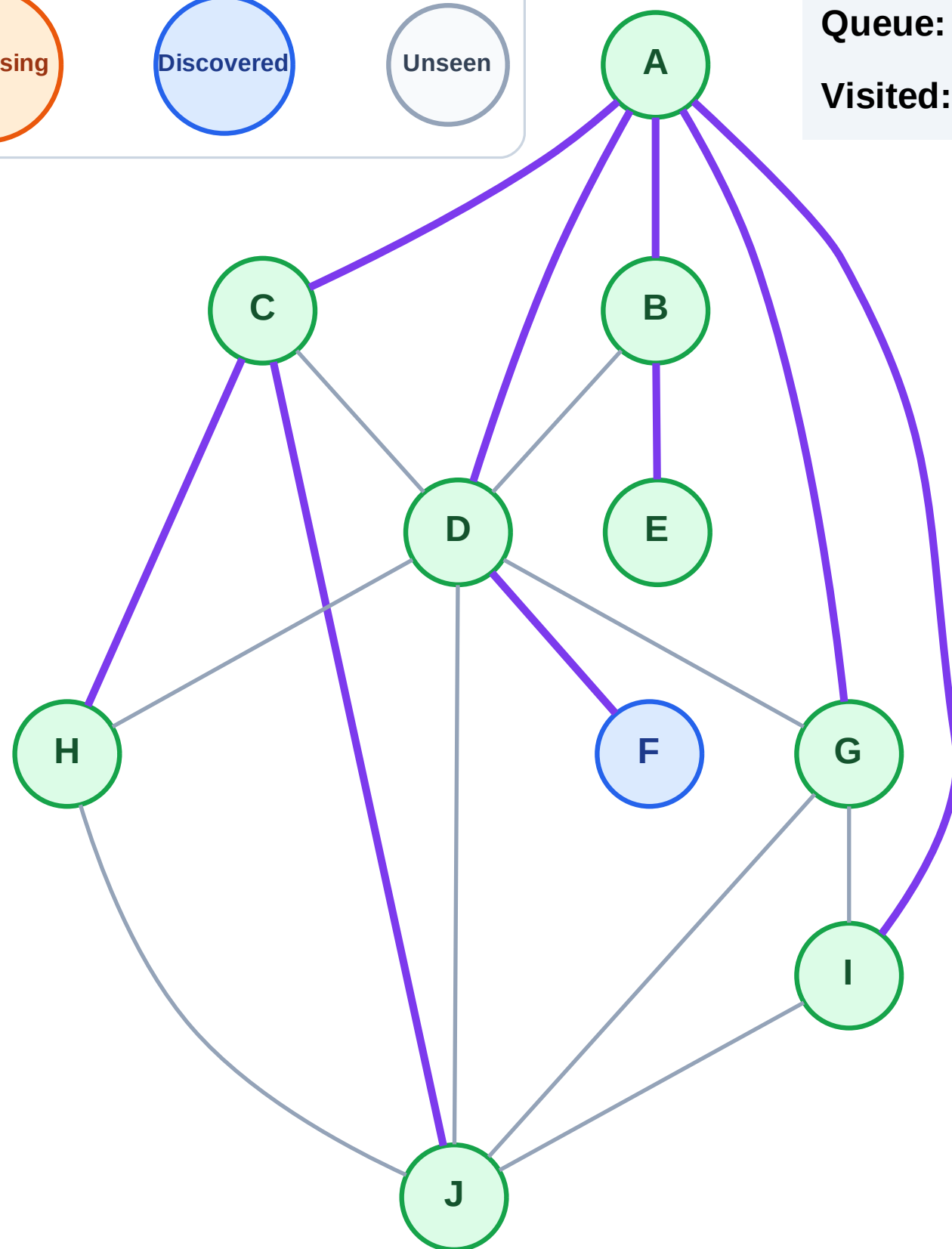
Step 19: No new neighbors from J

Legend



Queue: F

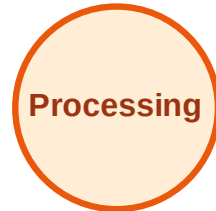
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

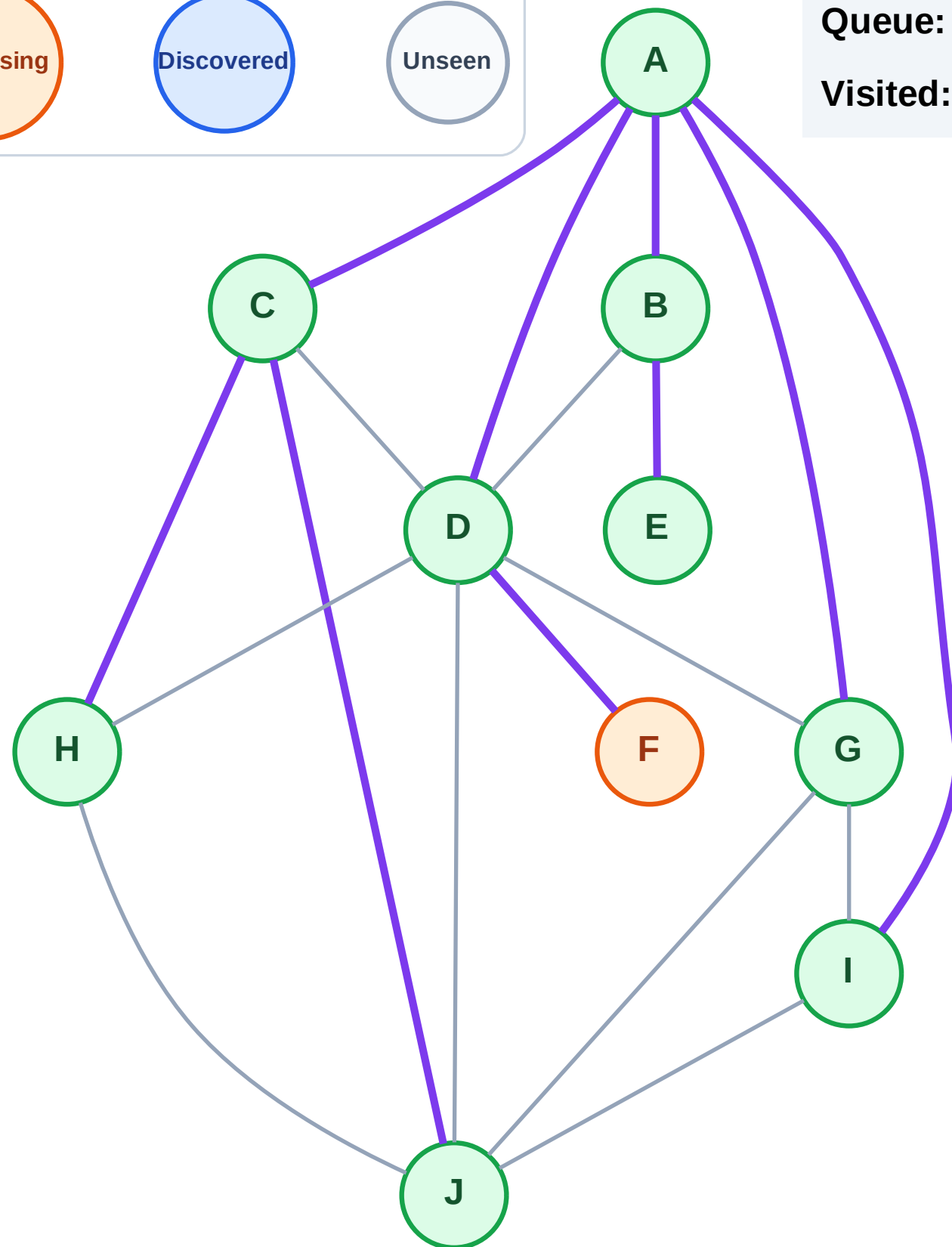
Step 20: Process node F

Legend



Queue: (empty)

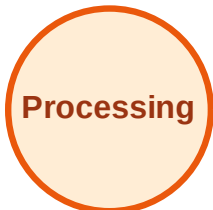
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

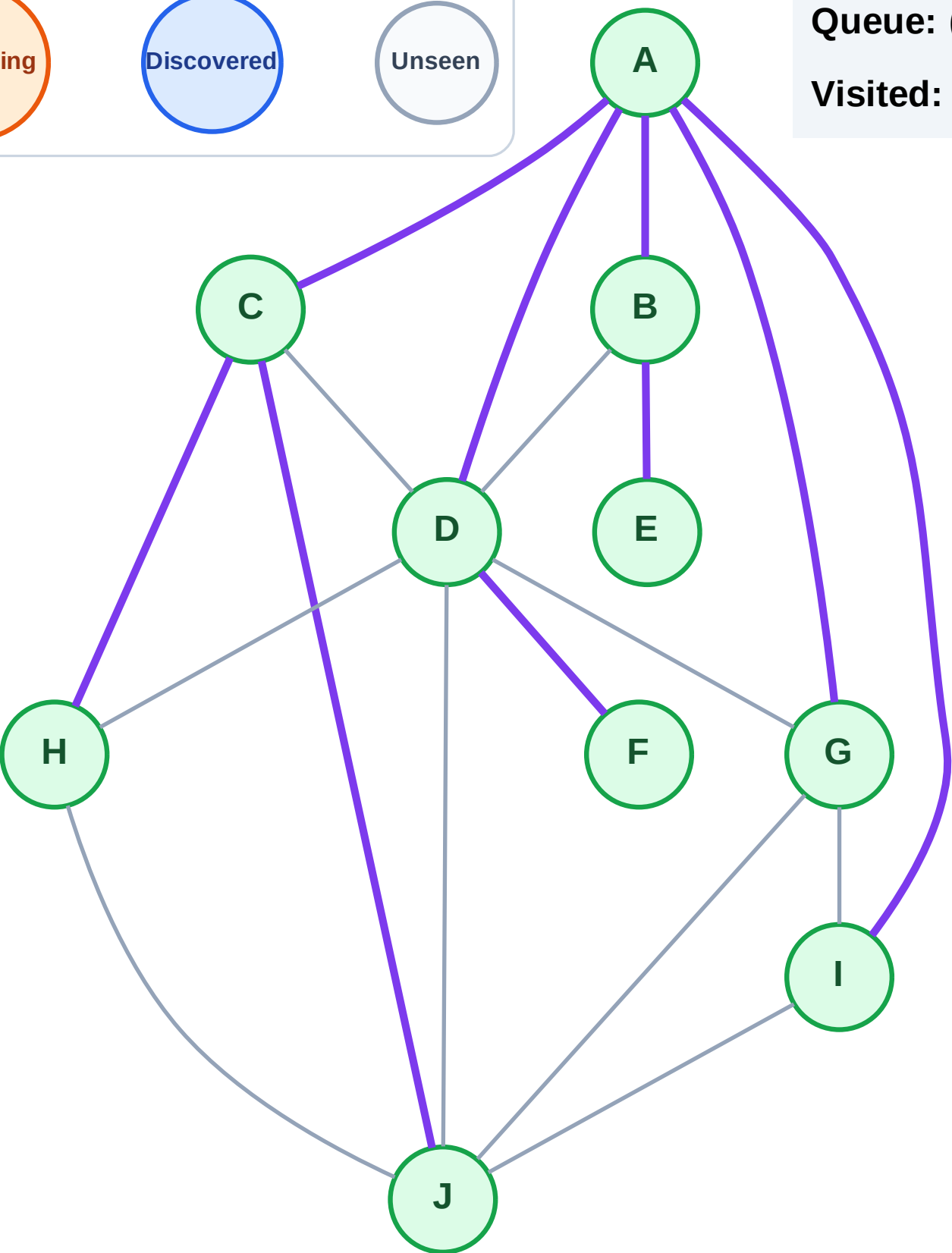
Step 21: No new neighbors from F

Legend



Queue: (empty)

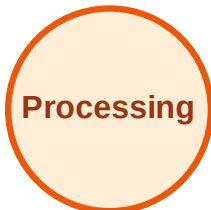
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

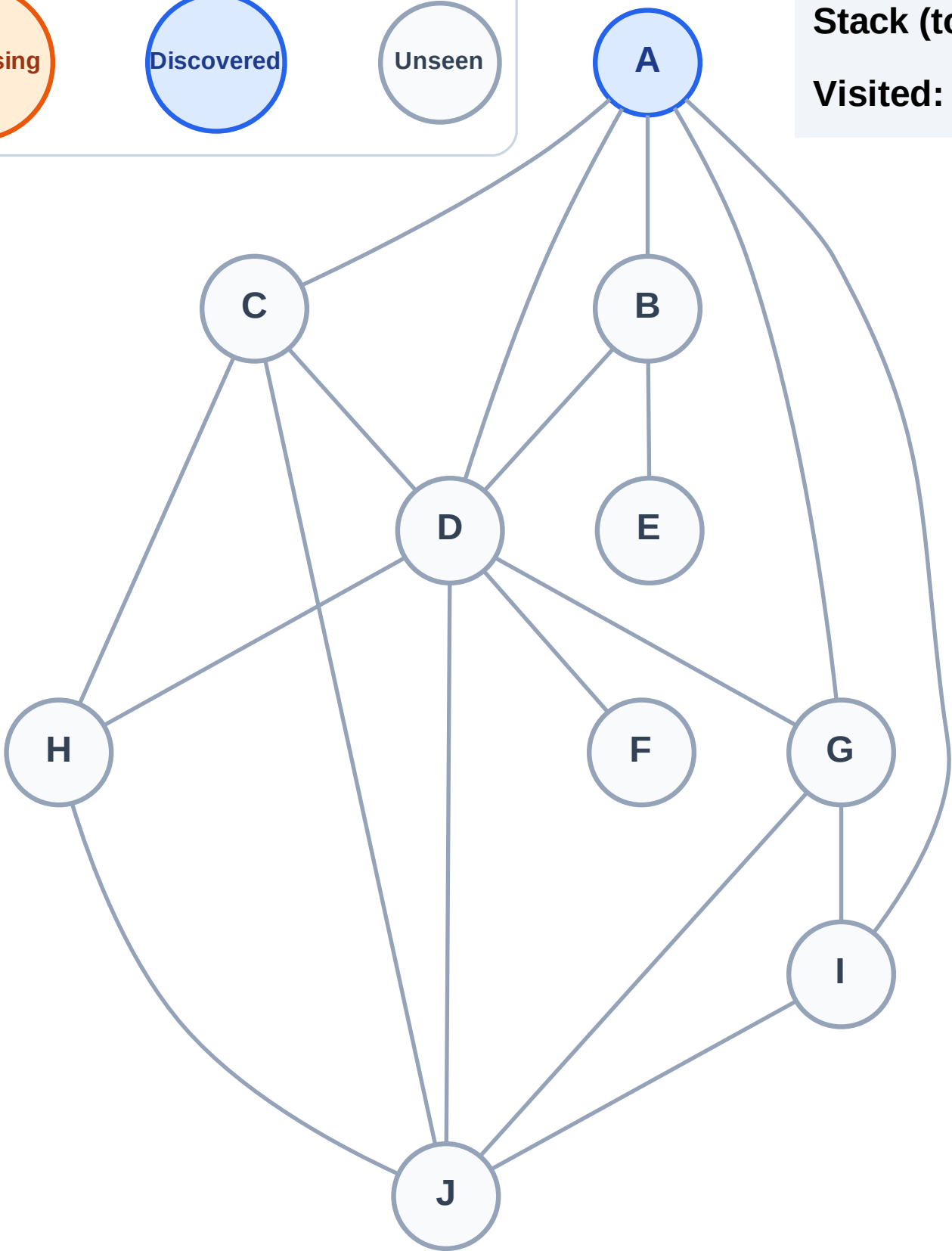
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

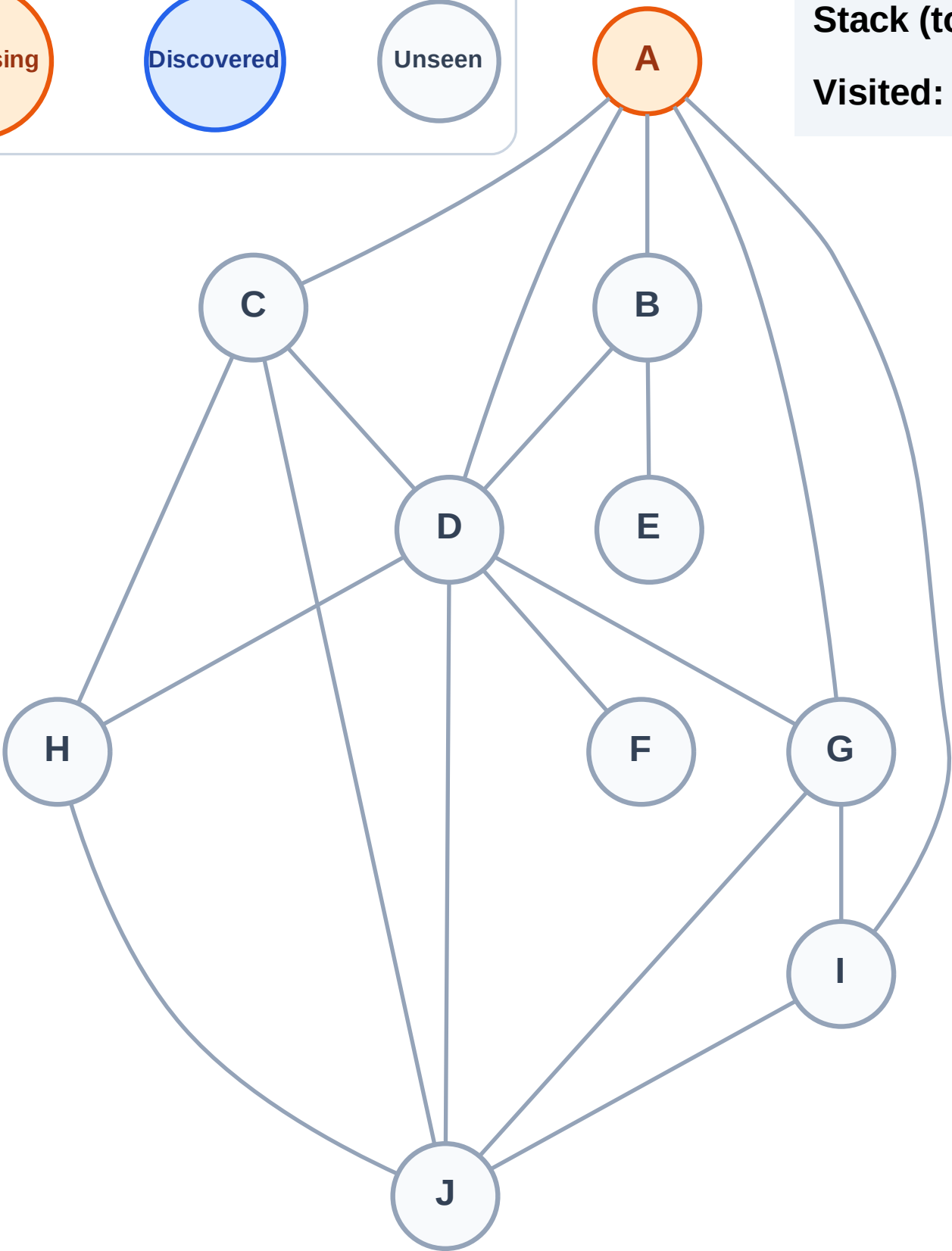
Complete

Processing

Discovered

Unseen

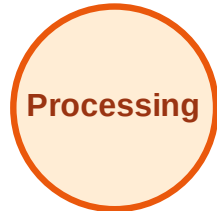
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

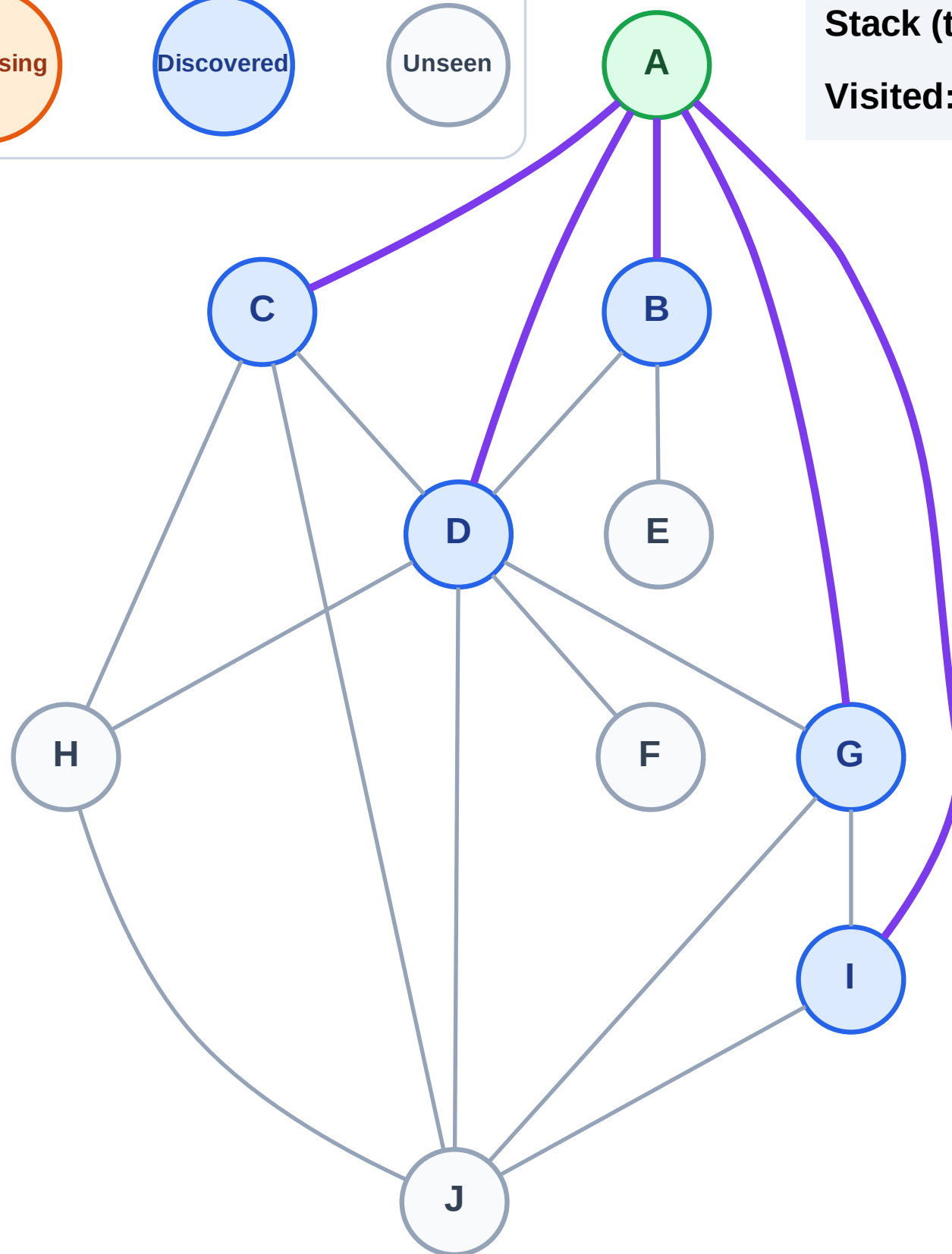
Step 3: Discover I, G, D, C, B from A

Legend



Stack (top at right): I | G | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

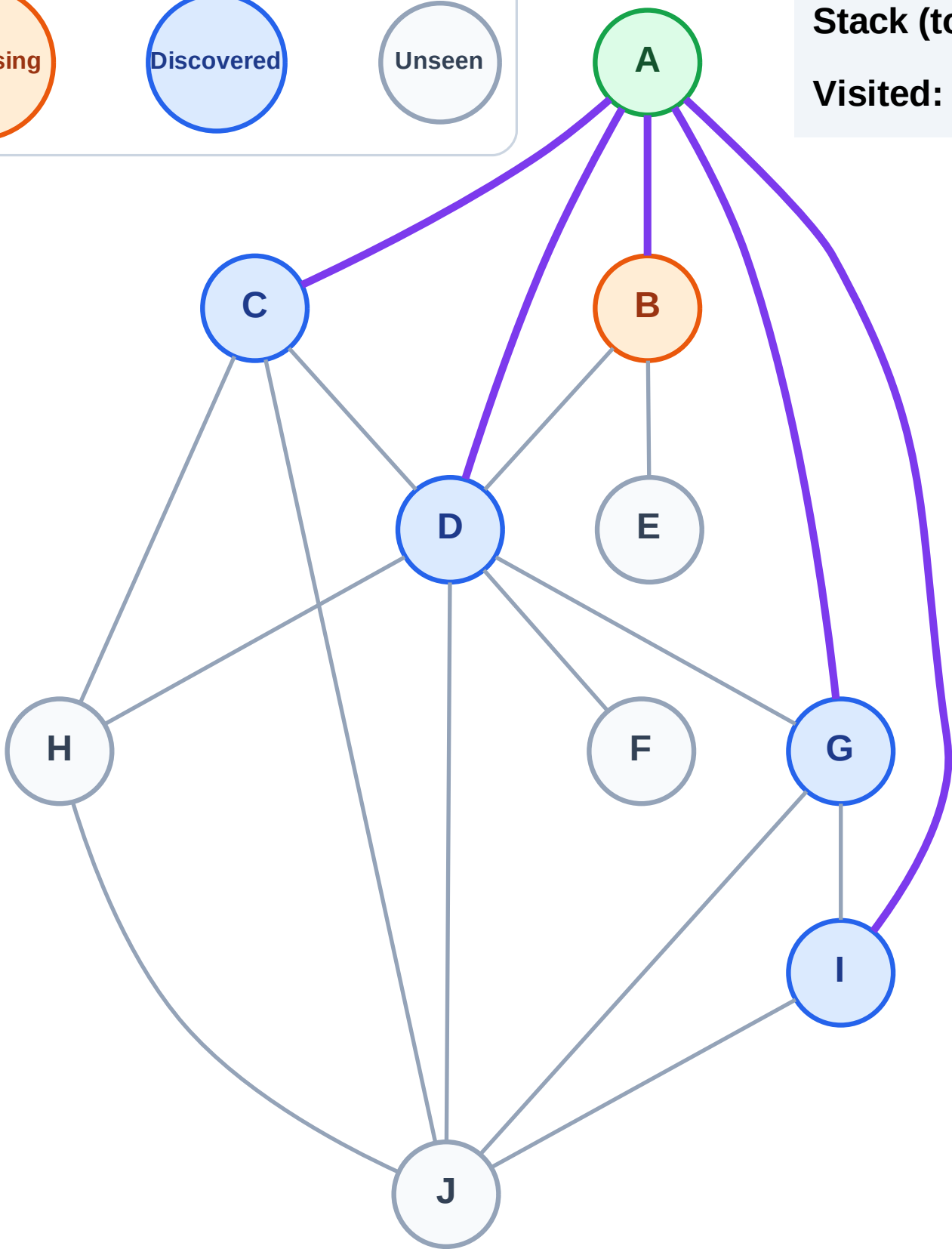
Complete

Processing

Discovered

Unseen

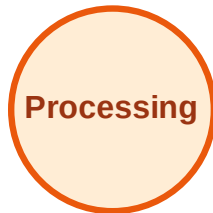
Stack (top at right): I | G | D | C
Visited: A



DFS Traversal

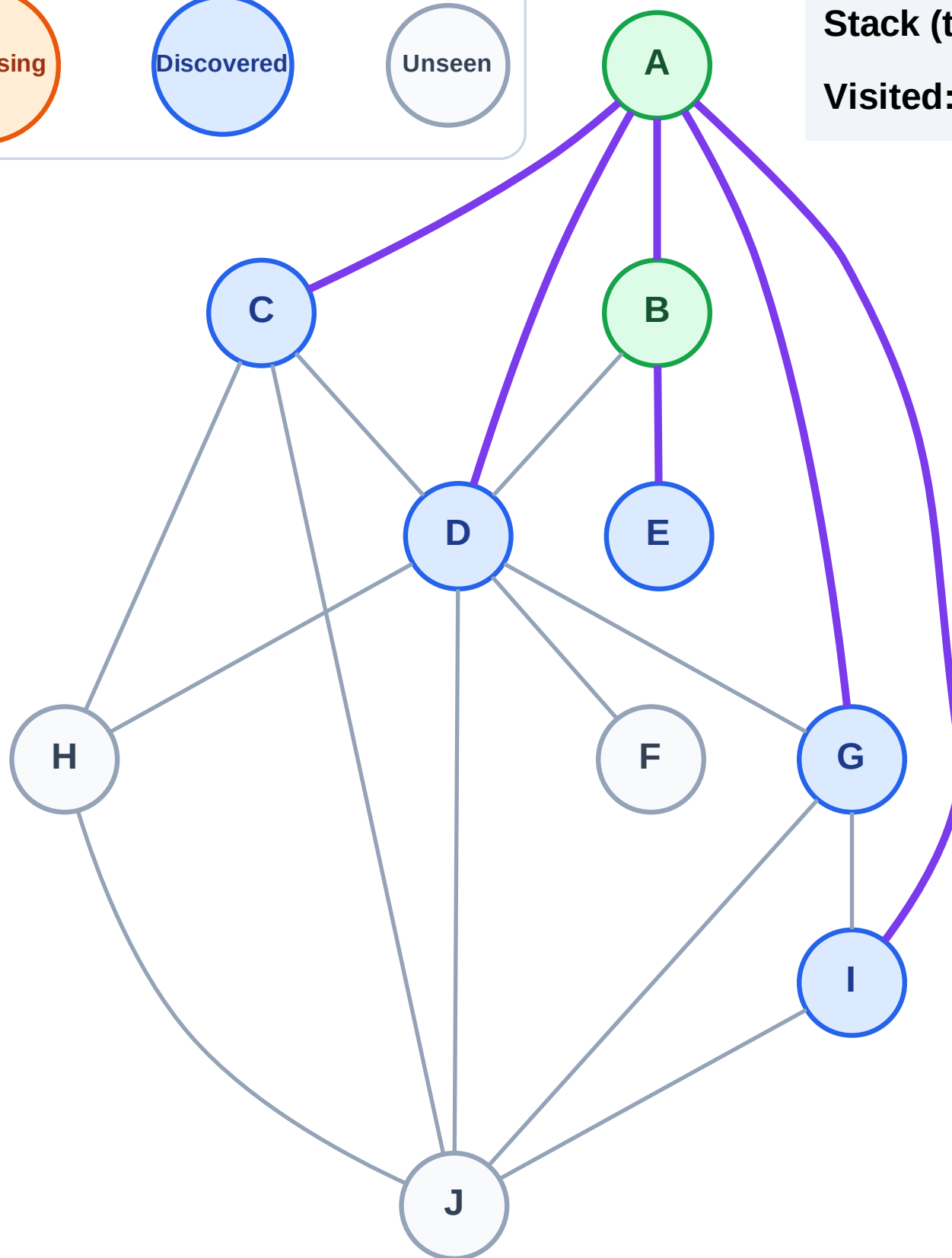
Step 5: Discover E from B

Legend



Stack (top at right): I | G | D | C | E

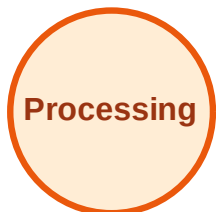
Visited: A, B



DFS Traversal

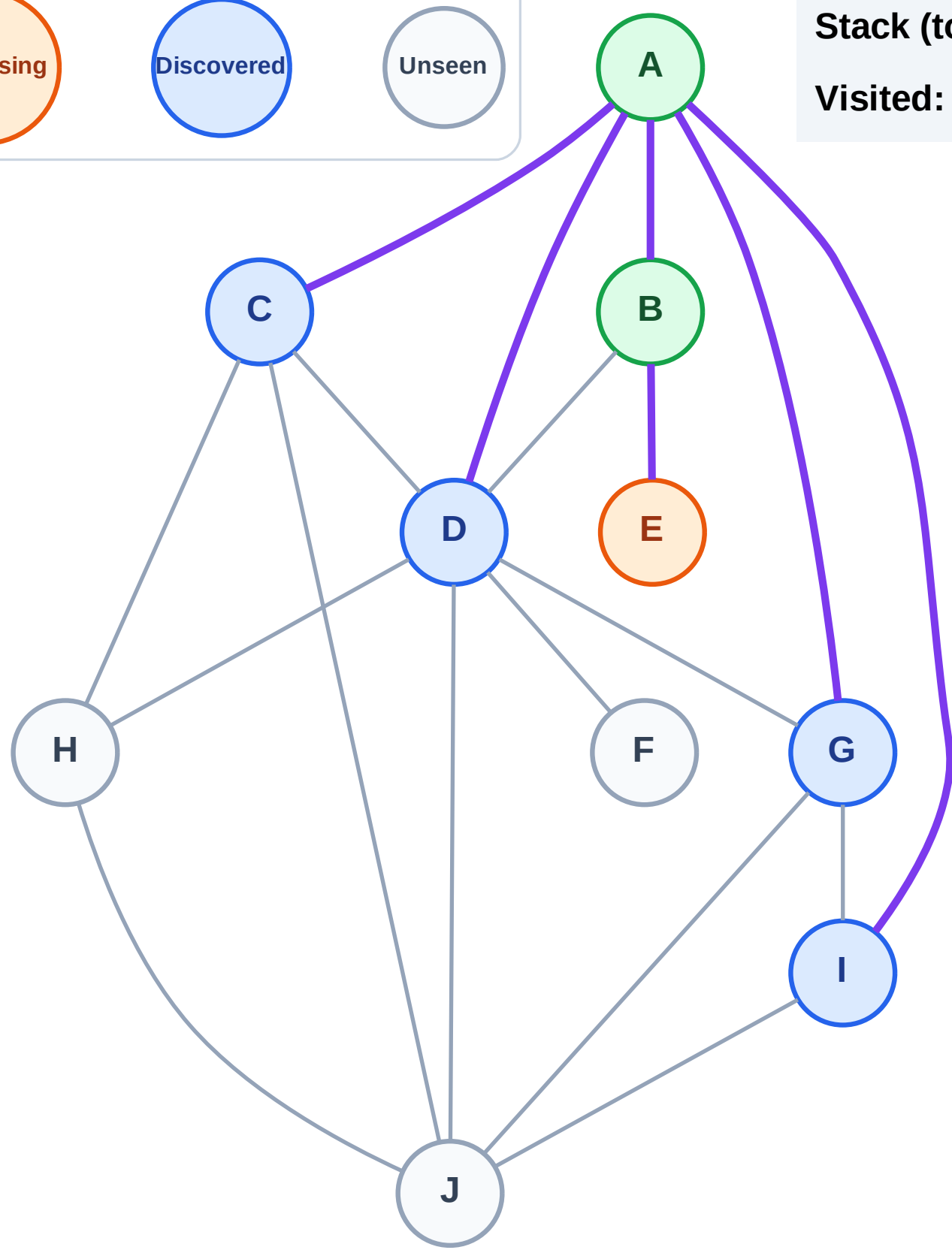
Step 6: Process node E

Legend



Stack (top at right): I | G | D | C

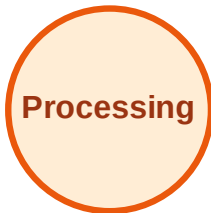
Visited: A, B



DFS Traversal

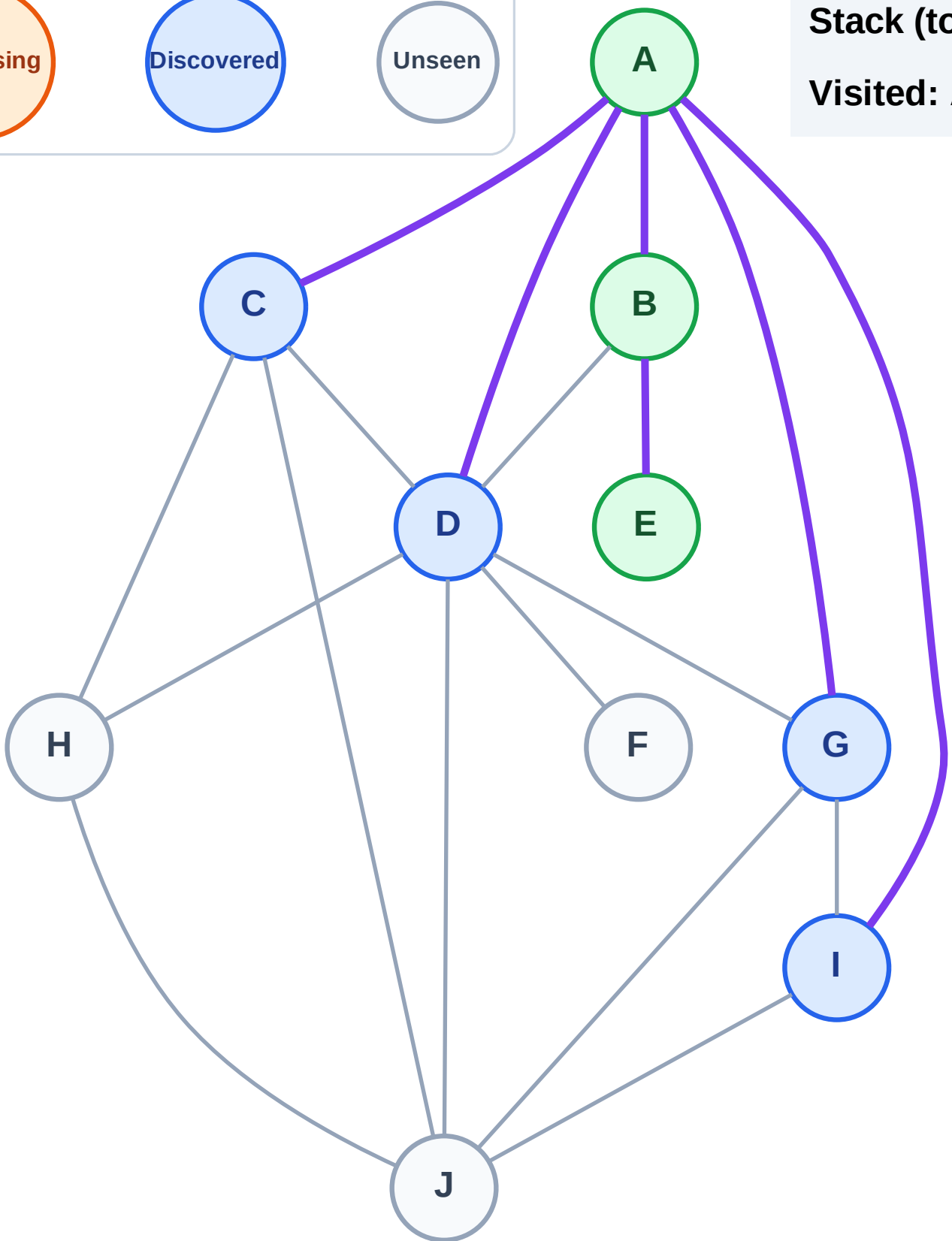
Step 7: No new neighbors from E

Legend



Stack (top at right): I | G | D | C

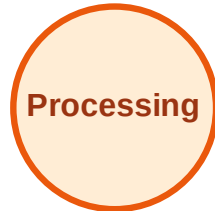
Visited: A, B, E



DFS Traversal

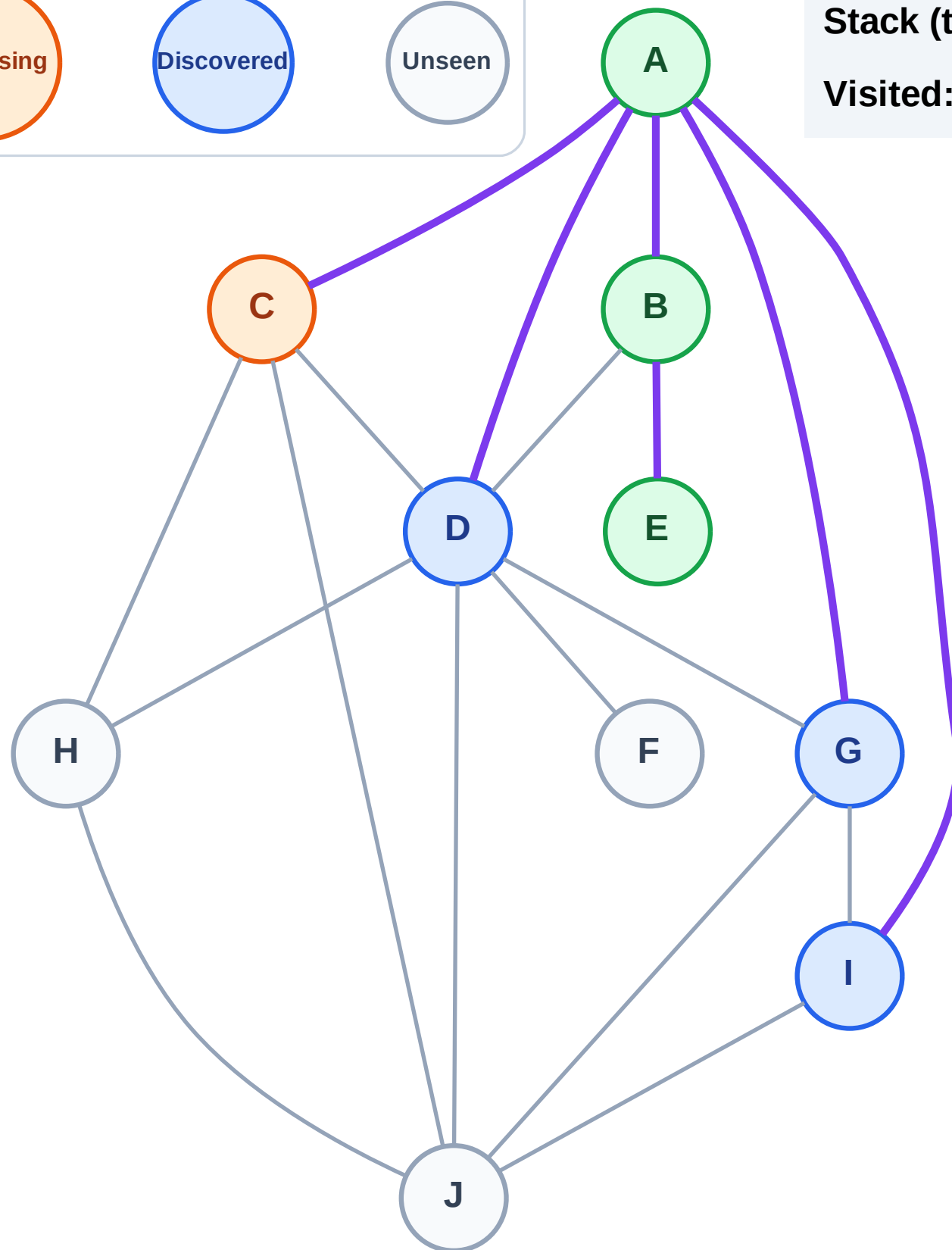
Step 8: Process node C

Legend



Stack (top at right): I | G | D

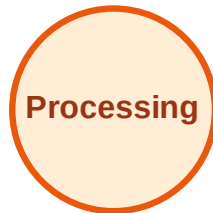
Visited: A, B, E



DFS Traversal

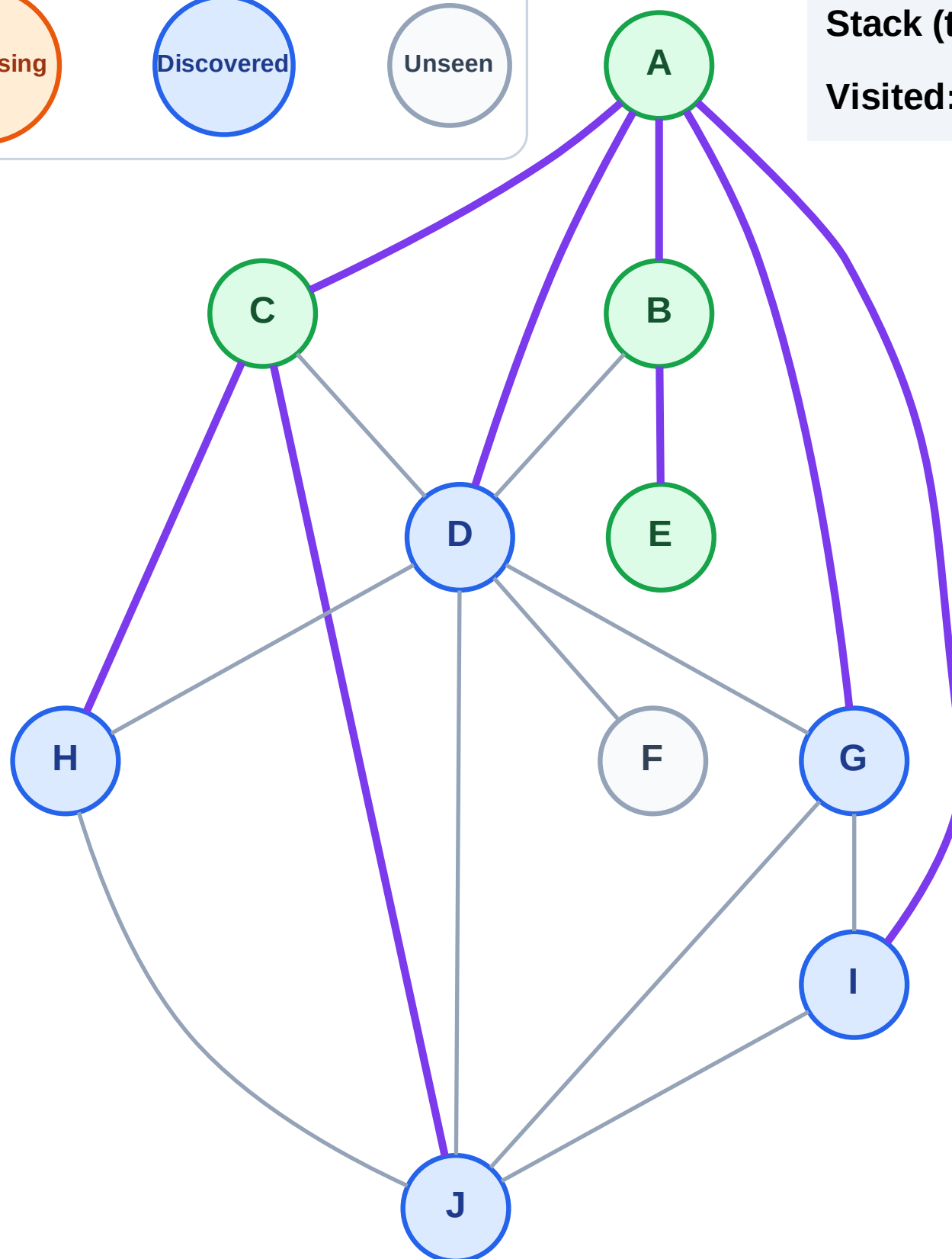
Step 9: Discover J, H from C

Legend



Stack (top at right): I | G | D | J | H

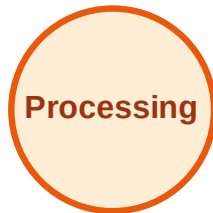
Visited: A, B, C, E



DFS Traversal

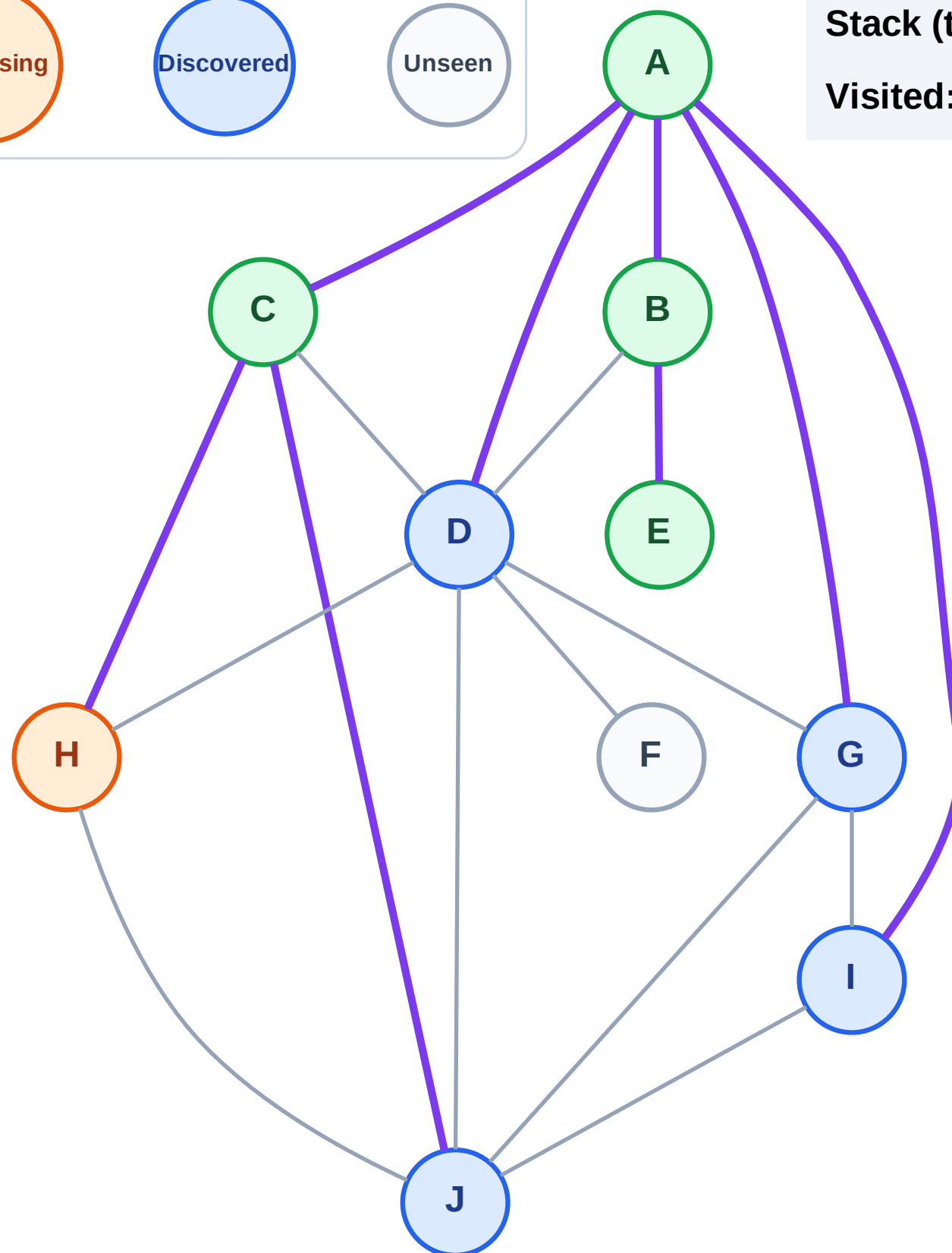
Step 10: Process node H

Legend



Stack (top at right): I | G | D | J

Visited: A, B, C, E



DFS Traversal

Step 11: No new neighbors from H

Legend

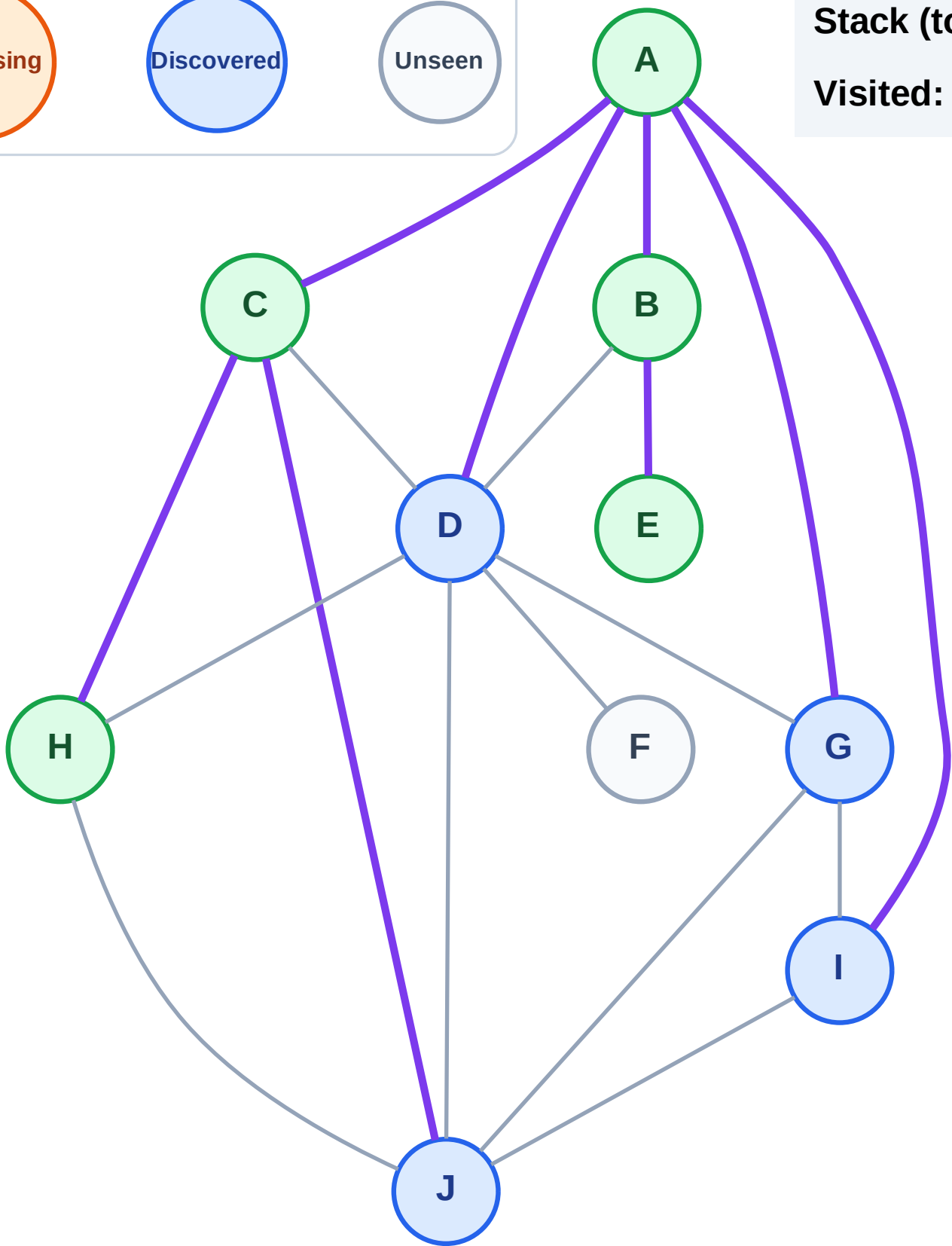
Complete

Processing

Discovered

Unseen

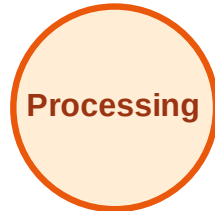
Stack (top at right): I | G | D | J
Visited: A, B, C, E, H



DFS Traversal

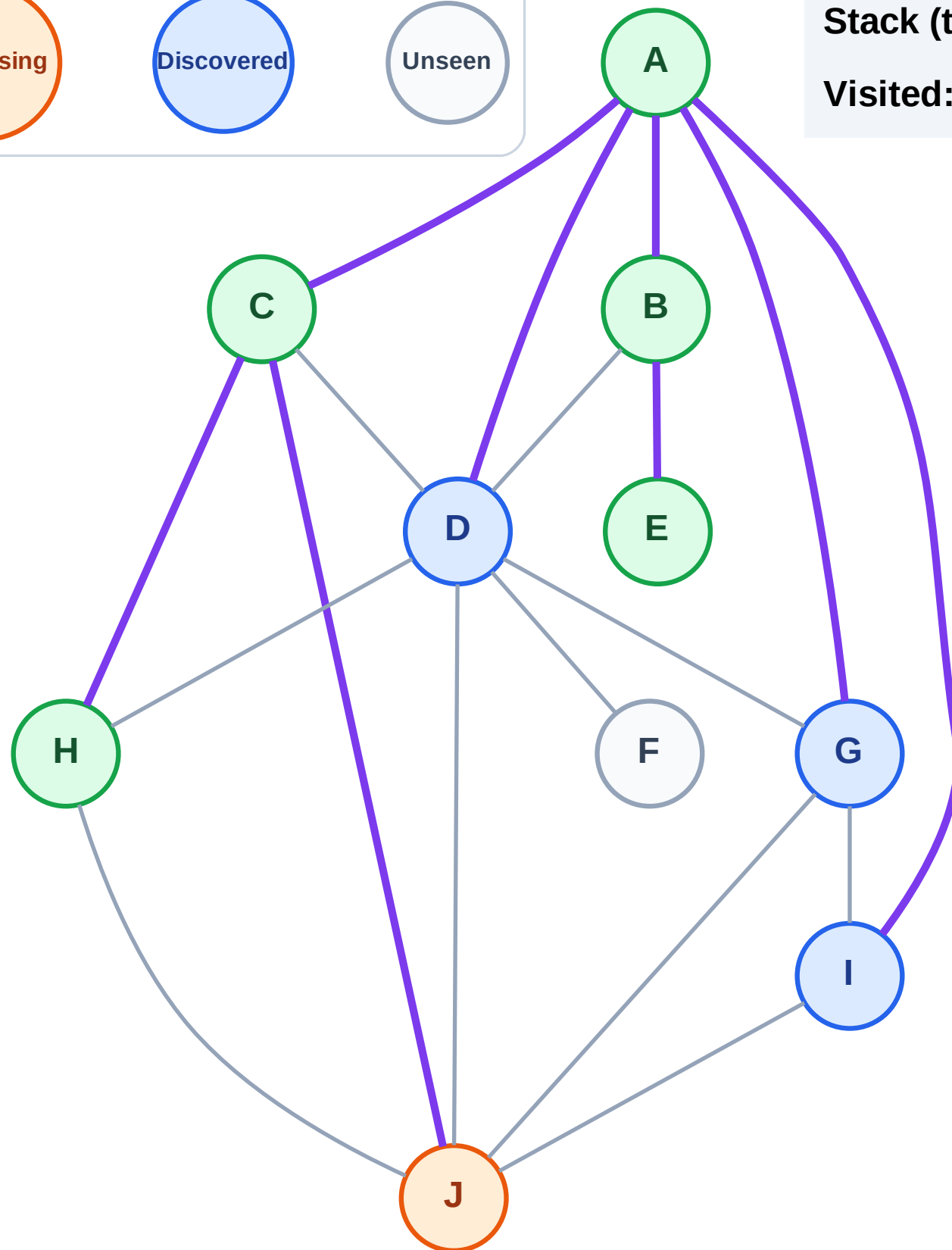
Step 12: Process node J

Legend



Stack (top at right): I | G | D

Visited: A, B, C, E, H

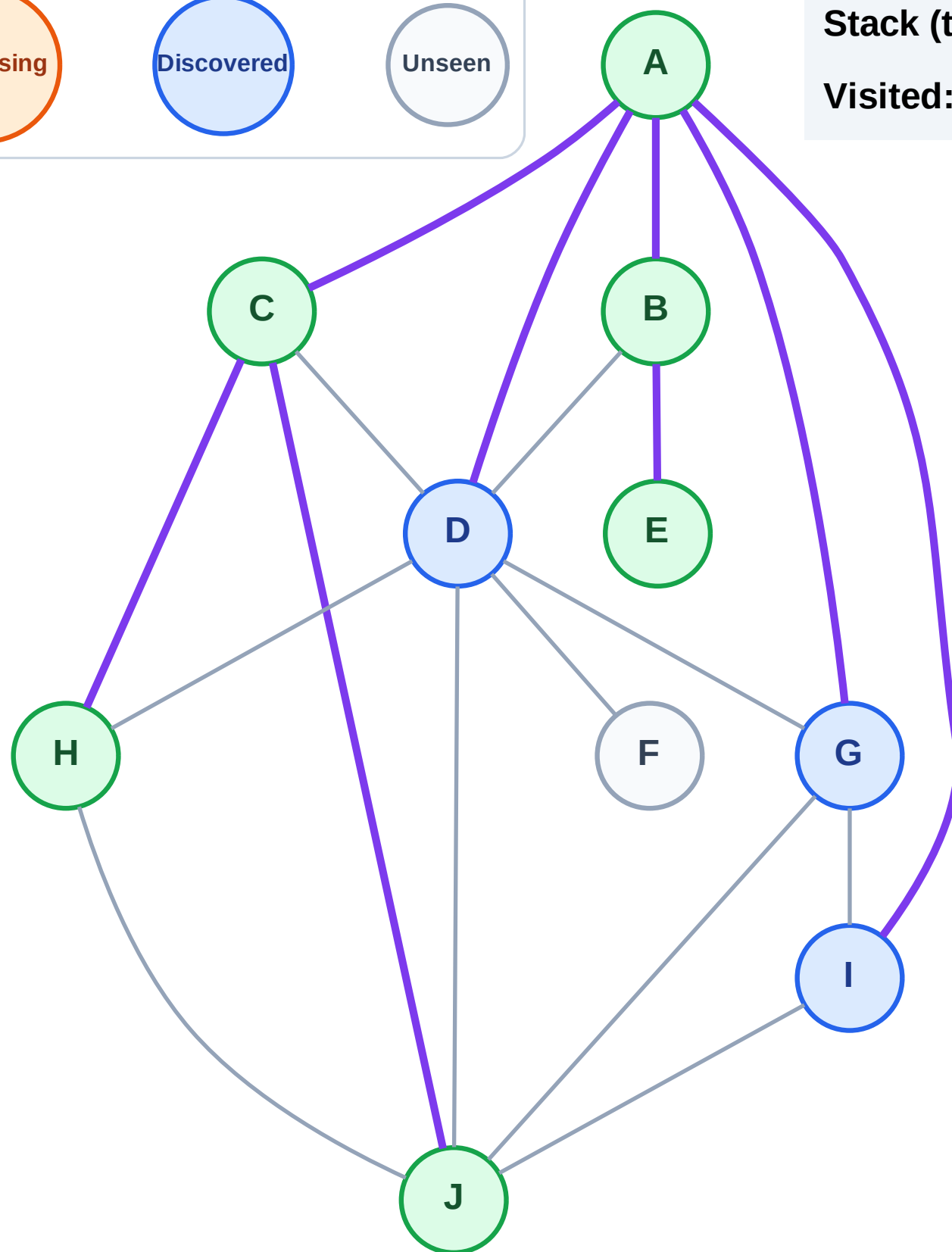


Step 13: No new neighbors from J

Legend

Unseen

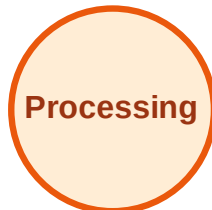
Visited: A, B, C, E, H, J



DFS Traversal

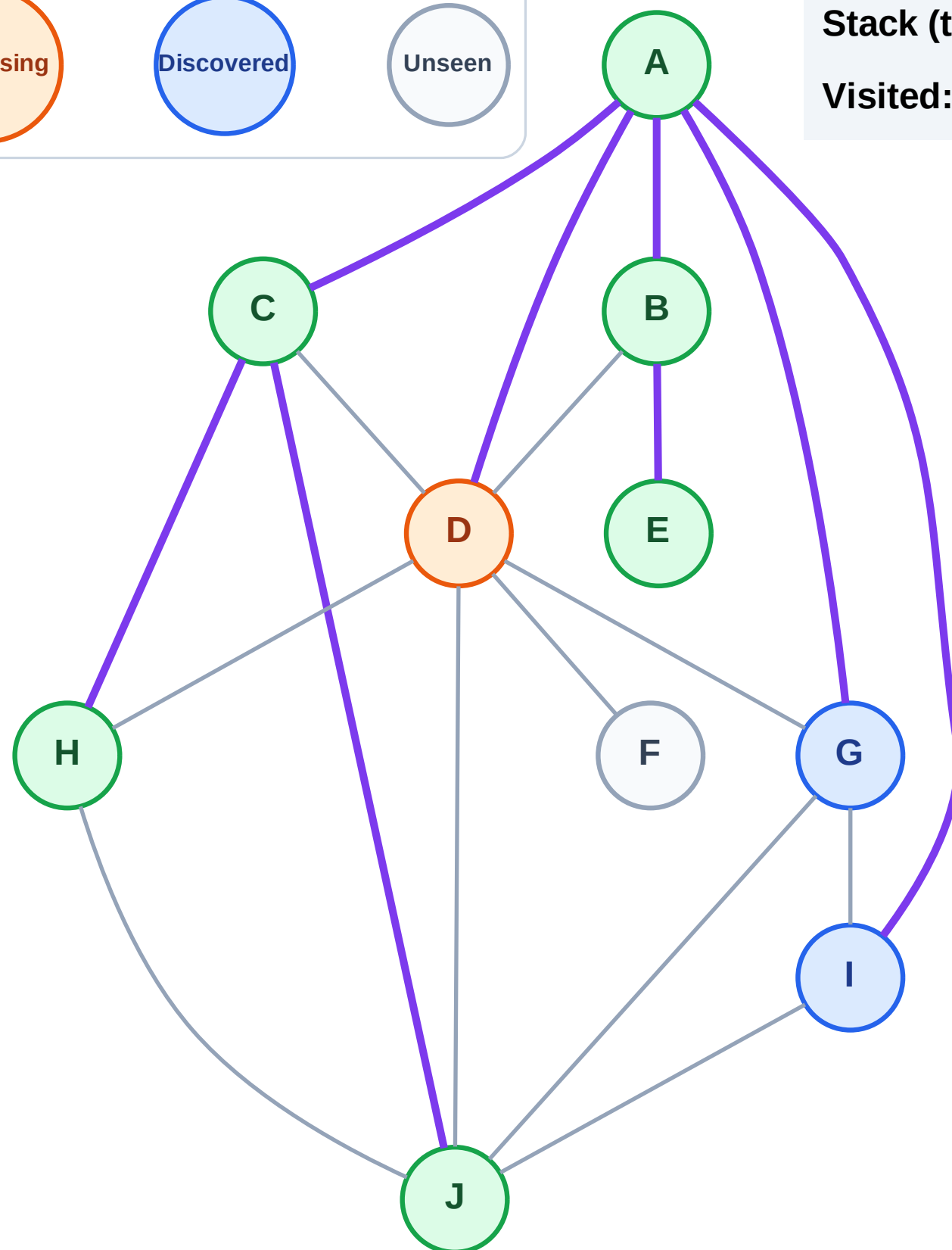
Step 14: Process node D

Legend



Stack (top at right): I | G

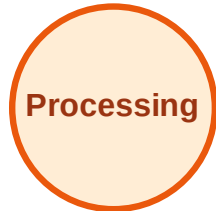
Visited: A, B, C, E, H, J



DFS Traversal

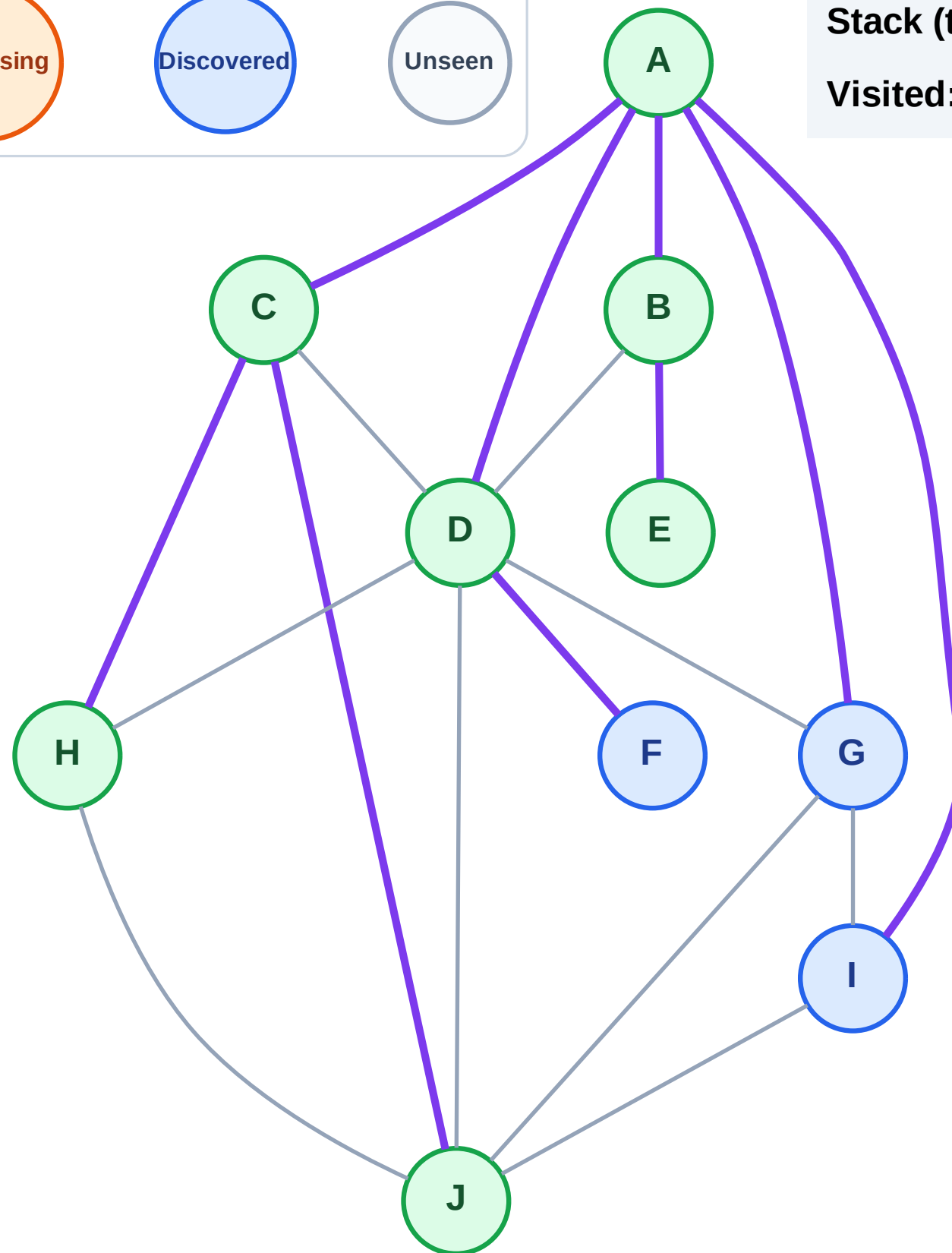
Step 15: Discover F from D

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

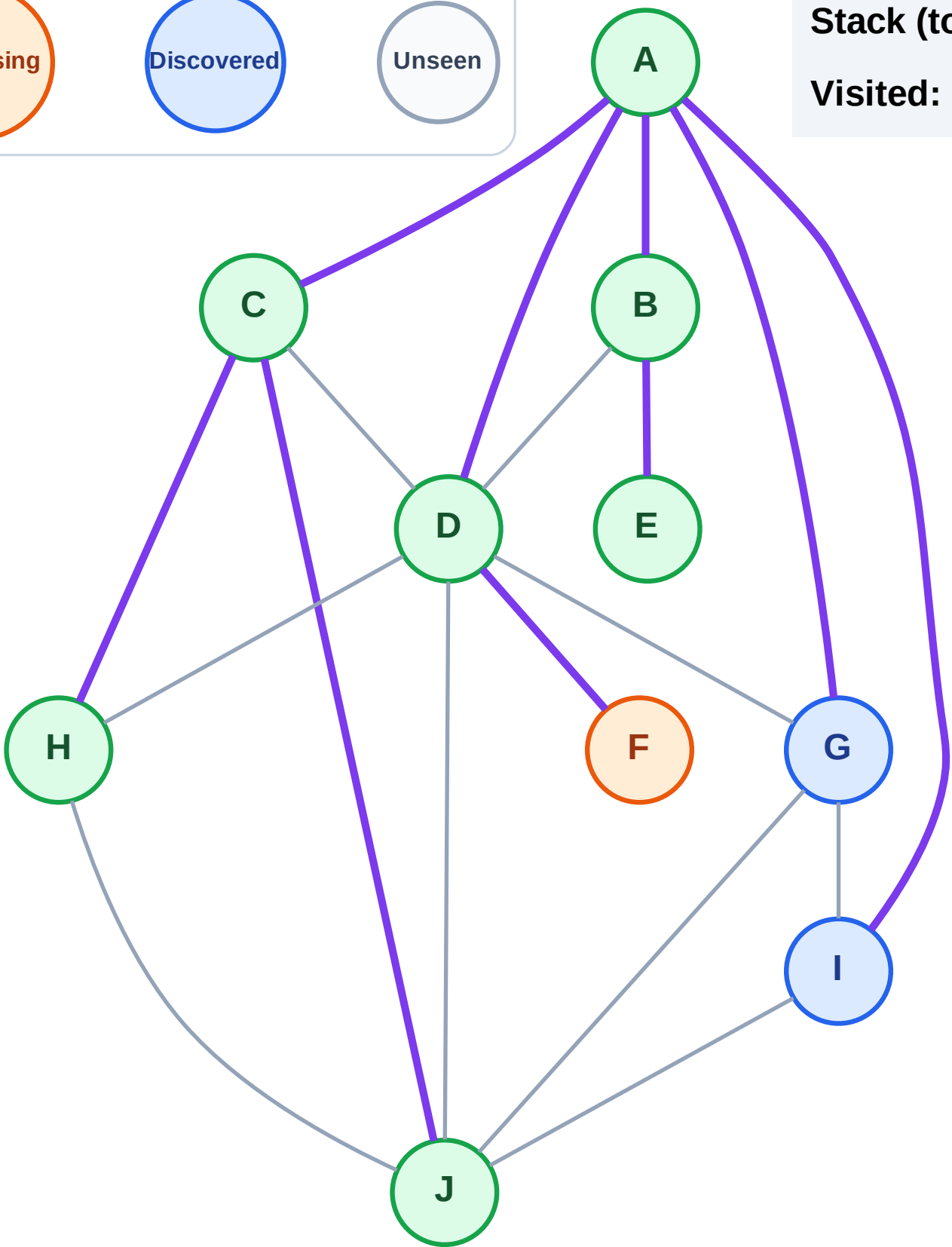
Complete

Processing

Discovered

Unseen

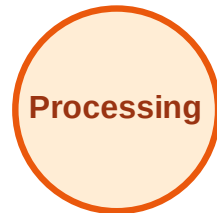
Stack (top at right): I | G
Visited: A, B, C, D, E, H, J



DFS Traversal

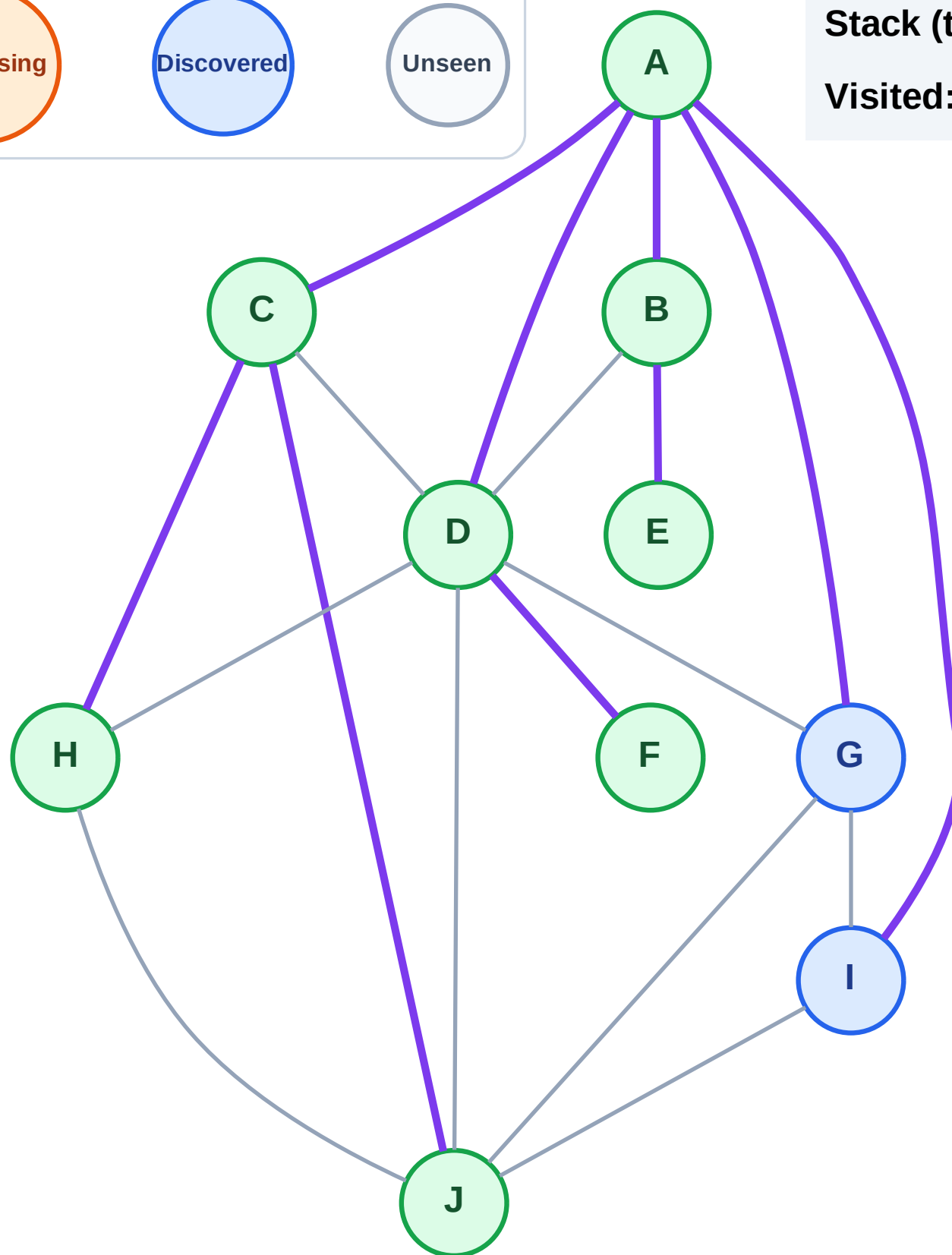
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

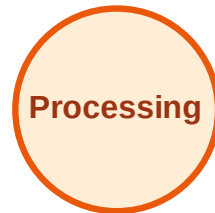
Visited: A, B, C, D, E, F, H, J



DFS Traversal

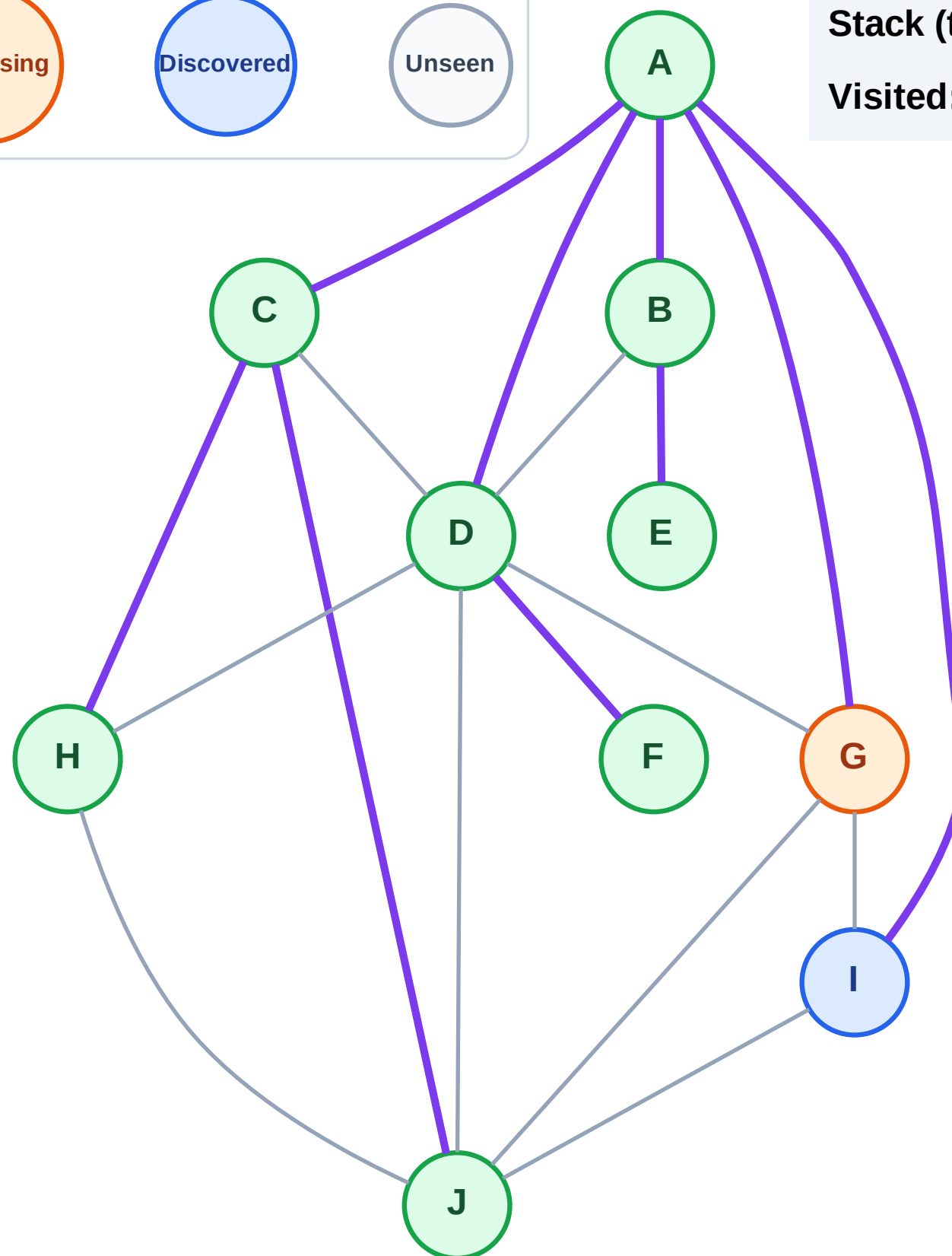
Step 18: Process node G

Legend



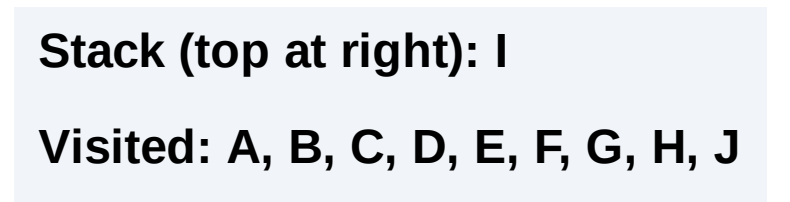
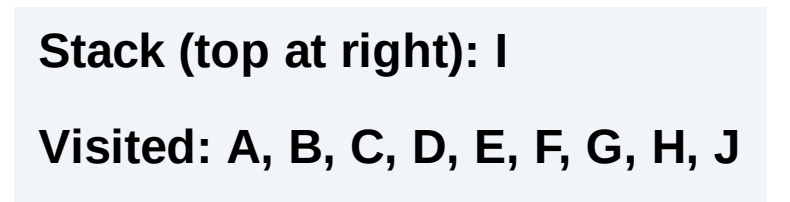
Stack (top at right): I

Visited: A, B, C, D, E, F, H, J



Step 19: No new neighbors from G

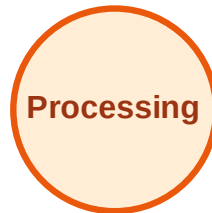
Step 19: No new neighbors from G



DFS Traversal

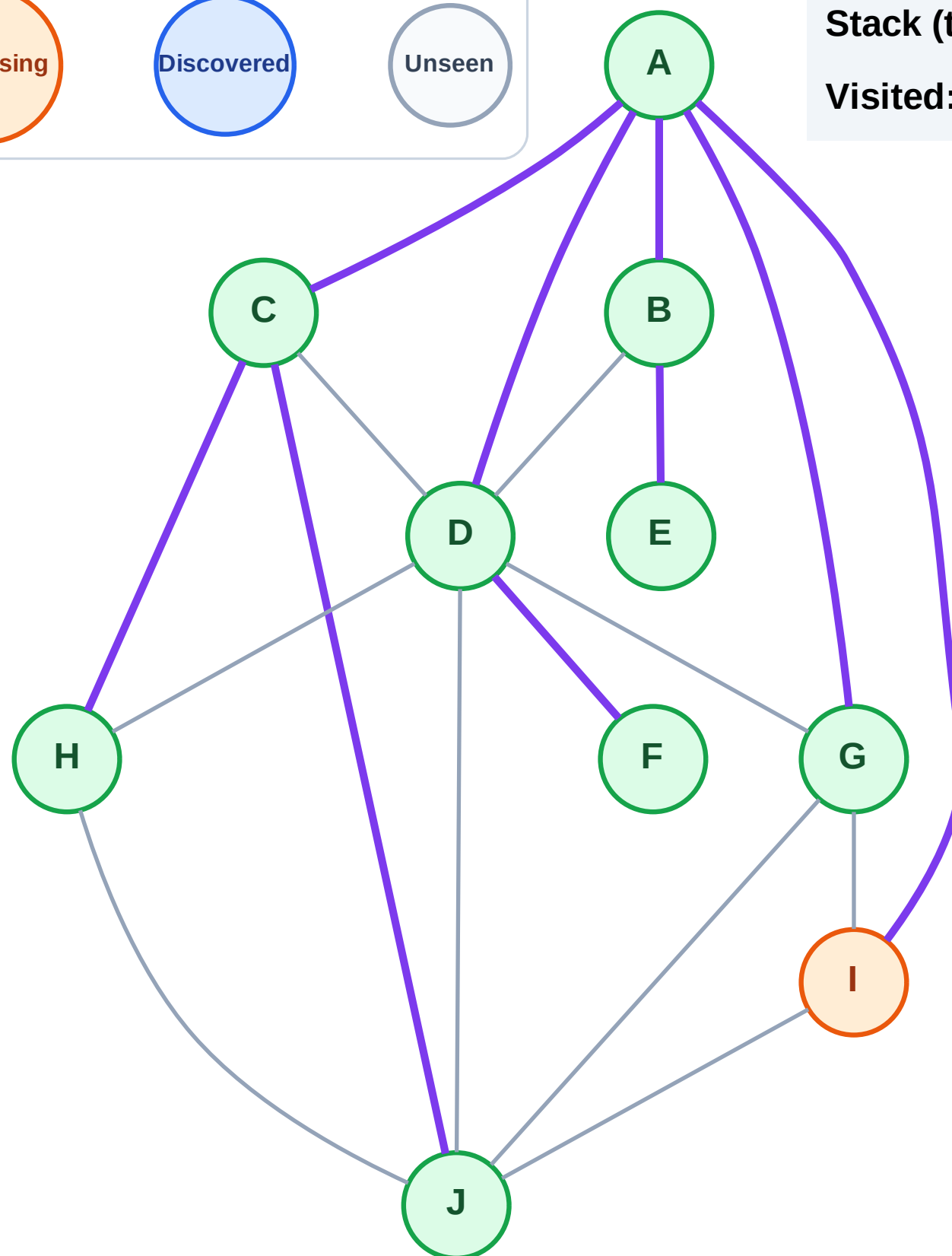
Step 20: Process node I

Legend



Stack (top at right): (empty)

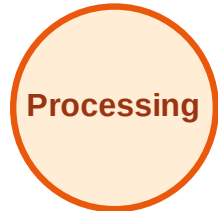
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

